

POPULATION SHOCKS, MIGRATION FLOWS, AND PERSISTENT REGIONAL DYNAMICS ^{*}

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Abstract

What are the local and aggregate dynamic effects of regional population pressure, and how do they evolve over time through endogenous migration responses? To address this question, I combine a panel SVAR framework with an external shift-share instrument that leverages lagged birth-rate variation to estimate the short-, medium-, and long-term effects of a 1% increase in migration inflows to U.S. commuting zones. The resulting impulse responses show that inflows raise output per worker and wages, with output per worker peaking around 0.2% above pre-shock levels and wages rising more gradually. At the same time, housing prices—an indicator of congestion—respond strongly and persistently, peaking around 0.35% and settling at a higher long-run level. I then embed these local dynamic causal effects in a quantitative dynamic spatial model with migration frictions and productivity spillovers, estimated by impulse-response matching. The model translates shocks that change regional population stocks into shifting congestion and agglomeration forces, evolving local attractiveness, and adjustments in gross inflows and outflows. Using counterfactual experiments calibrated to the geographic distribution of undocumented workers, I study region-specific labor removal and redistribution. Labor removal reduces aggregate output per capita by about 1% in the long run, while redistribution leaves the steady state unchanged but can generate temporary output losses of up to 1% and recovery times exceeding 50 years, depending on the skill composition of relocated workers.

Keywords: Migration, Labor Mobility, Economic Growth

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1 Introduction

What are the dynamic effects of local population shocks, including policy-driven relocations and demographic change, on local economies and the nation as a whole? Recent swings in internal migration that concentrate inflows in a subset of places have made clear that the time path of adjustment shapes employment opportunities, housing affordability, and the welfare consequences of policies that move people across space. Congestion forces, such as rents, respond quickly and can erode local attractiveness, while agglomeration forces that raise productivity may accrue only gradually; as these forces evolve, they shift local attractiveness and induce dynamic adjustments in gross inflows and outflows over time. However, much of the spatial literature abstracts from these transition dynamics, providing limited evidence on the dynamic causal response of rents, wages, and productivity to population pressure. Moreover, without a general-equilibrium mapping of how flows reallocate across regions, reduced-form responses alone cannot deliver aggregate counterfactuals when outcomes depend on which places receive people.

This paper aims to fill this gap. First, I estimate the dynamic local effects of migration shocks using a panel SVAR at the commuting-zone (CZ) level, coupled with an external instrument. The resulting structural impulse response functions (IRFs) trace how local wages, output, and housing costs evolve over time following a rise in population inflows. Second, I develop a quantitative dynamic spatial model with migration frictions and productivity spillovers, and use the empirical IRFs to discipline the model’s key elasticities. These parameters govern how population pressure shifts congestion and agglomeration, how those forces change local attractiveness, and how gross inflows and outflows adjust in response. With the estimated model in hand, I study aggregate outcomes in counterfactual experiments that change the spatial distribution of labor over time. In particular, I first consider a spatially heterogeneous labor-removal shock calibrated to regional undocumented-worker shares. I then implement a redistribution shock that relocates the same workers to other areas within the United States, allowing migration flows to respond endogenously as local attractiveness evolves.

To identify the local causal effect of migration inflows, I estimate a panel SVAR-IV model using shift-share instruments derived from lagged birth rates. This vector autoregression structure enables me to capture the intertemporal relationships between population inflows and other key variables. As a result, I can account not only for the direct impact of migration inflows on economic outcomes but also for the potential feedback effect, where favorable economic conditions attract additional future migration. To address potential simultaneity bias arising from contemporaneous productivity shocks that could drive migration into the area, I employ an instrumental variable approach. After estimating the reduced-form panel VAR, I apply the methodology of [Stock and Watson \(2018\)](#) to identify the impact of structural market size shocks using an external instrument. Specifically, for each CZ,

I predict population inflows using a Bartik-style instrument: the 'pull' factor are lagged migration shares at the destination and the 'push' factor is the lagged birth rate of each origin, following a similar approach to [Karahan, Pugsley and Şahin \(2019\)](#).

The exclusion restriction of this instrument relies on the assumption that birth rates in the areas sending migrants are uncorrelated with destination-specific shocks 25 years later. Following the strategy used in the immigration literature, such as in [Altonji and Card \(1991\)](#) and [Card \(2001\)](#), the weights of the shift-share instrument correspond to the lagged share of migrants originating from each area. However, unlike most studies in this field, I use migration patterns from the previous year rather than a fixed pre-period. Although these shares may be endogenous, I follow [Borusyak, Hull and Jaravel \(2022\)](#) and estimate the IV at the 'shock level'—in this case, the level of the county sending migrants. This approach allows me to leverage the exogeneity of past birth rates while constructing a stronger instrument with inflow shares that more accurately predict current migration inflows. Additionally, it enables me to calculate the appropriate standard errors for inference, thereby avoiding the limitations of conventional shift-share methods discussed by [Adao, Kolesár and Morales \(2019\)](#).

At the CZ level, a 1% increase in migration inflows leads to persistently higher future immigration, raises wages and output, reduces net establishment growth, and drives up housing prices. The shock produces an immediate effect on all variables, and several responses build over the next few years. Specifically, output per worker rises on impact by about 0.14% and peaks around 0.20% after three years, while average wages increase more gradually to roughly 0.10% after four years. At the same time, some workers begin to leave the area, as out-migration rises by 0.55% after three years, consistent with higher housing prices that increase by 0.34% after four years. However, these cumulative effects then ease slightly from their peaks and stabilize at higher levels: for example, after peaking at 0.37% housing eventually settles near 0.25% above pre-shock values by year ten, while net establishment entry turns negative after about three years and remains depressed thereafter.

I also provide additional evidence to shed light on the mechanisms behind the baseline results. First, I add other measures of local dynamism that show the decline in establishment net-entry is driven primarily by lower gross firm entry and a sustained fall in establishments per worker. Furthermore, job destruction falls sharply and persistently while job creation remains mostly unchanged. This suggests adjustment occurs mainly through larger incumbent firms rather than extensive-margin entry. Second, I split CZs by population density (population per square mile) and re-estimate the SVAR-IV separately for each group. Housing prices are the only outcome that differs meaningfully across groups: prices rise earlier and more persistently in larger, denser markets, peaking at 0.6–0.7% by year three and remaining elevated thereafter. Smaller or less dense areas show lower and more delayed increases that even fade out over time. Third, because the instrument is positively correlated

with migrants’ income per capita, the modest rise in average wages could partly reflect composition. To address this, I examine wage responses within narrowly defined industries with relatively similar skill requirements. I find substantial sectoral heterogeneity: wages rise in some industries, including trucking and transportation, but fall in others, consistent with inflows increasing local demand for certain services while expanding labor supply more strongly in other sectors.

These results underscore why it is essential to estimate the dynamic effects of migration inflow shocks. First, the impulse responses show that the impact is not concentrated on arrival: several outcomes continue to build or decline over subsequent years, reflecting slower adjustment mechanisms such as housing construction, changes in firm entry, and additional migration inflows (and outflows) in subsequent years. Second, the dynamic specification helps account for persistent, unobserved local shocks by conditioning on lagged outcomes, which would otherwise contaminate the IV estimates. Finally, estimating the time profile of flow shocks helps discipline the transition following shocks to local population stocks: when policy or demographic forces change the number of residents in a place, congestion and agglomeration incentives shift, local attractiveness changes, and gross inflows and outflows adjust over time, shaping both local and aggregate outcomes.

Given these results, how does aggregate productivity react to local population shocks? Inherently, this question connects heterogeneous treatments at the local level to aggregate outcomes and answering it requires a theoretical framework. I therefore develop a continuous time spatial economic model in which forward looking workers choose where to live each period, subject to mobility costs on the intensive and extensive margin. Furthermore, the model includes static agglomeration externalities à la [Romer \(1990\)](#) as well as local dynamic spillovers similar to the ones in [Peters \(2022\)](#), where a larger local mass of firms in a region decreases the cost of entry in the future. On the other hand, endogenously congested amenities as well as a housing sector with fixed land make locations less attractive as population increases. Mobility frictions slow the spatial redistribution of labor, and, when combined with intertemporal productivity spillovers, create conditions in which migration shocks can have persistent impacts that vary over time on variables like wages and housing prices. This is what we observe in the empirical IRFs.

To discipline the main parameters of the model I use impulse response matching. In particular, I simulate the transition path of the economy implied by the empirical birth rates in the shift-share instrument. I then replicate the estimation described above using the simulated data to obtain model generated IRFs. The parameter estimates I obtain are the ones that minimize the distance between these simulated moments and the empirical values. Because the same parameters govern how population pressure affects congestion and agglomeration—and therefore local attractiveness—this procedure also matches the migration elasticities and adjustment speeds that determine how shocks to population stocks propa-

gate through endogenous migration flows. Other parameters are estimated using calibration by targeting cross-sectional evidence with moments generated in the initial steady-state.

I then use the calibrated model to assess the aggregate impact of local population shocks. To do so, I construct two policy counterfactuals. The first is a “deportation” shock: using the empirical geographical distribution of undocumented migrants in the U.S., this scenario removes a segment of the workforce from the economy. It is essential to emphasize that these are region-specific shocks, with deportation shares varying across regions, representing between 3% and 5% of local populations. The areas directly affected by the shock experience higher amenities, lower housing prices, and real incomes that are, on average, 7.5% higher. As a result, these locations attract workers from other regions, with increased inflows and reduced outflows. However, the overall reduction in the workforce leads to a long-term decline in the number of firms and total output by 6% and 2%, respectively, across all regions. These dynamics ultimately cause aggregate output per capita to decline to a new steady state that is 1% below the initial level. The magnitude of these effects depends on the skill composition of the deported workers, with potentially greater impacts if higher-skilled workers are removed.

In contrast, the second policy scenario involves a “relocation” shock, in which the same population is redistributed uniformly across other regions. A similar policy has been implemented recently at a local level, with some states relocating undocumented immigrants to different regions within the country (Goodman et al. (2024)). The results in this paper indicate that, if applied on a national scale, such a policy would produce no long-term effects, as the economy would eventually converge back to the original steady state through migration flows. However, in the short term, the shock would create significant impacts, including a 1% increase in housing prices across all locations and an immediate 9% rise in real incomes in areas losing workers. Moreover, the reduction in the mass of firms by more than 2% in these regions would lead to a regional output decline of between 2% and 8%, depending on the skill composition of the relocated households. At the aggregate level, relocating low-skilled households results in a slight increase in output per capita. Conversely, redistributing high-skilled workers leads to a drop in output per worker of over 1%, with recovery taking more than 50 years.

Related literature This paper follows previous research on market size, labor mobility, and economic growth. From a theoretical perspective, Duranton and Puga (2004) summarize the main mechanisms driving agglomeration economies. The first of these is *sharing*: producers in larger cities can divide the fixed cost of indivisible inputs as well as the gains of specialization. Another channel is *matching*, as a larger labor force increases both the quantity and quality of firm-worker matches, especially in the presence of mismatching costs due to skill heterogeneity. Finally *learning* from a larger and more diverse market raises productivity through knowledge creation, diffusion, and accumulation (Lucas, 1988;

Moretti, 2004; Crews, 2023). The relevance of each mechanism, and ultimately the effect of immigration, depends on the time frame. For example, a fixed capital stock in the short run causes wages to drop in the immediate years following worker inflows (Borjas, 2013). By estimating a dynamic model over several years, the following analysis determines how the relative importance of each mechanism evolves.

Numerous empirical studies estimate the response to labor inflows and obtain ambiguous results. Using previous immigration shares as an instrument, some papers find a negative effect on native employment and wages (Altonji and Card, 1991; Card, 2001). Similarly, the wage elasticities estimated by Borjas (2003) imply immigration to the US in the 1980s and 1990s reduced wages by 3.2%. Nevertheless, as the survey by Edo (2019) points out, the estimated impact on wages, employment, and other outcomes depends on the methodology, period, and country under study. Some papers have found evidence of agglomeration economies with positive spillovers on manufacturing productivity (Greenstone, Hornbeck and Moretti, 2010; Kline and Moretti, 2014) as well as employment opportunities (Moretti, 2010). Most of these studies estimate a single elasticity for multiple periods and adopt reduced-form strategies that do not account for the intertemporal correlation between migration and other economic measures.

To incorporate these interactions, this paper models local population inflows and outflows, along with other outcome variables, using a panel VAR structure. Barcellos (2010) adopted this approach in her analysis of migration between US states whereas Boubtane, Coulibaly and Rault (2013) did so for 22 OECD countries. To identify the dynamic response to population inflow shocks, these studies rely on ordering assumptions and Cholesky decompositions of the variance-covariance matrix. I avoid making such assumptions by estimating the coefficients of interest using an IV strategy as described by Stock and Watson (2018) and applied in the fiscal multipliers (Mertens and Ravn, 2013) as well as the monetary policy literature (Gertler and Karadi, 2015).

The use of exogenous variation resembles the analysis of previous authors leveraging historical episodes to estimate the effect of population movements. Notable examples include Peters (2022) and Burchardi and Hassan (2013), both of whom examine the impact of post-WWII German migrants who resettled in West Germany. However, their estimation relies on a single event, whereas the SVAR-IV exploits changes across time. To accomplish this, I use a shift-share instrument (estimated at the shock level as stated in Borusyak, Hull and Jaravel (2022)) based on migration shares and lagged birthrates. This follows a similar identification strategy as the one proposed by previous work that estimates the effect of increasing labor supply (Shimer, 2001; Karahan, Pugsley and Şahin, 2019).

The methodology in this paper identifies exogenous shocks that influence local immigration flows to estimate the latter’s effect on economic conditions. In this sense, it closely resembles the Local Projection IV (LP-IV) used in Howard (2020), which uses a

dynamic model along with an [Altonji and Card \(1991\)](#)-style instrument to determine the effect of migration on unemployment. The paper finds that unemployment falls whereas the employment-to-population ratio rises. However, it differs from the analysis I carry out in several ways. First, its econometric approach does not account for the interaction between migration, unemployment, and other relevant variables such as wages, output, and establishments, which is an element the VAR structure allows me to consider. Furthermore, it only estimates the effect on labor market outcomes, whereas this paper determines the response on a set of comprehensive economic measures. Finally, I estimate the impact on CZs which cover a broader geographical sample than MSAs, which are only representative of urban economies.

This paper also follows previous work on the implications of local dynamics for aggregate outcomes. [Davis, Fisher and Whited \(2014\)](#) study the contribution of agglomeration in cities to aggregate productivity growth through the lens of a structurally estimated spatial model. In their framework, local output per worker depends positively on output density. [Crews \(2023\)](#) further endogenizes these externalities in a heterogeneous agents model: as the aggregate stock of skill in a city increases, the rate of individual human capital accumulation rises as well¹. On the other hand, [Peters \(2022\)](#) proposes a different mechanism by relating current fixed costs of entry to past market size, creating dynamic spillovers. He embeds this in a model with variety gains, as in [Romer \(1990\)](#), and uses it to examine the scale effects on productivity in the context of postwar Germany.

The rest of the paper is structured as follows. The next section explains the SVAR-IV methodology, as well as the shock-level shift-share instrument I use to estimate the effects of migration. Then, in Section 3, I analyze the structural impulse responses to a 1% increase in inflows and discuss further supporting evidence. Section 4 develops the formal framework relating the local evidence to aggregate outcomes, whereas Section 5 implements the structural estimation of the model parameters. Finally Section 6 implements the policy counterfactuals and Section 7 concludes.

2 Econometric Model

2.1 Panel SVAR-IV Model

The first part of this paper seeks to determine the dynamic effect of population inflows on local economies. Specifically, for any CZ i and year t , I determine the impact total in-migration from within the US m_{it} has on the population flow dynamics in the years following the initial shock. In other words, I estimate the effect on future in- and out-migration, m_{it} and o_{it} respectively, since the arrival of new residents could drive native

¹Previous papers in the growth literature suggested this mechanism ([Uzawa, 1965](#); [Lucas, 1988](#)), albeit at an aggregate level in a representative agent economy.

workers to leave as well as more immigrants to relocate in the future. This could be due, in part, to changes in employment opportunities. Consequently, I also study the effect on the employment-to-population ratio l_{it} to establish if new workers compete for the same jobs as previous employees or if their arrival leads to additional opportunities.

Another relevant labor market outcome is the average wage w_{it} within a CZ, as an in-migration shock can increase labor supply and could therefore lead to lower wages. On the contrary, the presence of agglomeration economies can make labor productivity increase with larger market size. These two opposing effects can lead to different net impacts across different timelines. In addition to wages, I also estimate the effect on output per worker y_{it} , another measure of aggregate productivity, as well as on the establishment net entry rate k_{it} . Through the lens of an expanding variety model, a larger workforce leads to higher entry, which in turn increases productivity. Thus, accounting for establishments will allow me to test the predictions of such frameworks.

Finally, I determine the response of housing prices hp_{it} , a major component of households' local expenditures. As the population of an area increases, these values should increase in the short run, when the good is in fixed supply. This would in turn act as a congestion force, counteracting the agglomeration externalities and therefore leading to different effects across timelines. Moreover, recent trends in real estate prices, especially in markets experiencing a high inflow of migrants, have prompted a policy discussion aimed at increasing the housing supply. Consequently, to study these counterfactual policies we first need to estimate the effect of population increases on real estate values.

To estimate these effects, the first step is to model these variables using a panel SVAR structure, similar to the one in [Boubtane, Coulibaly and Rault \(2013\)](#), which accounts for the reciprocal relation between population flows and economic performance in a given geographical area. In this sense, define a vector of stationary observables $\mathbf{Y}_{it}$ and assume it follows a linear process²:

$$\mathbf{A} \cdot \mathbf{Y}_{it} = \sum_{l=1}^p \boldsymbol{\alpha}_l \cdot \mathbf{Y}_{it-l} + \boldsymbol{\varepsilon}_{it} \quad (1)$$

where $\mathbf{A}$ is a non-singular matrix and $\boldsymbol{\alpha}_l$ is a $7 \times 7p$ matrix which, along with $\mathbf{A}$, describes the relation between present and past values of $\mathbf{Y}_{it}$. Furthermore, $\boldsymbol{\varepsilon}_{it}$ is a vector of structural innovations uncorrelated across localities and time, with $E[\boldsymbol{\varepsilon}_{it}] = 0$ and $E[\boldsymbol{\varepsilon}_{it}\boldsymbol{\varepsilon}_{it}'] = I$. The reduced form of the model above is given by:

$$\mathbf{Y}_{it} = \sum_{l=1}^p \boldsymbol{\delta}_l \cdot \mathbf{Y}_{it-l} + \mathbf{B}\boldsymbol{\varepsilon}_{it} \quad (2)$$

where $\mathbf{B} = \mathbf{A}^{-1}$ and $\boldsymbol{\delta}_1 = \mathbf{A}^{-1}\boldsymbol{\alpha}_1$.

²This includes all the variables discussed above so that $\mathbf{Y}_{it} = [\Delta m_{it}, \Delta o_{it}, \Delta y_{it}, \Delta w_{it}, \Delta l_{it}, \Delta k_{it}, \Delta hp_{it}]'$.

External Instrument Let $\mathbf{u}_{it} = \mathbf{B}\boldsymbol{\varepsilon}_{it}$ be the reduced-form residuals. After estimating equation (2) we need to identify $\mathbf{B}$ to construct impulse responses to structural migration shocks ε_{it}^m . Typically, when making timing assumptions, we would rely on the Cholesky decomposition of Σ , the covariance matrix of $\mathbf{e}_{it}$, to achieve this. This is the approach [Boubtane, Coulibaly and Rault \(2013\)](#) adopt in their analysis of OECD immigration. However, since I am only interested in the response to immigration shocks, and given the ordering of variables in $\mathbf{Y}_{it}$, I only need to identify the first column $\mathbf{b}^m$. By instead using instrumental variables as detailed in [Stock and Watson \(2018\)](#), I avoid imposing any timing or sign restrictions.

To see this, rewrite the reduced-form residuals of the j^{th} VAR equation as a function of the structural migration ε_{it}^m and non-migration ε_{it}^{-m} shocks:

$$u_{it}^j = b_j^m \cdot \varepsilon_{it}^m + \mathbf{b}_{j,2:5}' \cdot \boldsymbol{\varepsilon}_{it}^{-m} \quad (3)$$

where b_j^m corresponds to the first coefficient in the j^{th} row of matrix $\mathbf{B}$. If we use a unit-effect normalization, so that $b_m^m = 1$, we can substitute ε_{it}^m with the residuals from the migration equation and obtain:

$$u_{it}^j = b_j^m \cdot u_{it}^m + e_{it}^j \quad (4)$$

where e_{it}^j is a linear combination of the non-migration structural shocks ε_{it}^{-m} .

As [Stock and Watson \(2018\)](#) note, although the residuals u_{it}^j are unobserved, to estimate b_j^m we can use each equation in the reduced-form VAR in (2) to rewrite equation (4) as³:

$$Y_{it}^j = b_j^m \cdot m_{it} + \sum_{l=1}^p \boldsymbol{\Gamma}_l^j \cdot \mathbf{Y}_{it-l} + e_{it}^j \quad (5)$$

where Y_{it}^j is the j^{th} entry in $\mathbf{Y}_{it}$. Since in-migration rates m_{it} are correlated with non-migration shocks we have $E[m_{it}^m e_{it}^j] \neq 0$. Consequently, estimating the coefficients of interest requires a vector of instruments $\mathbf{z}_{it}$ that satisfies the relevance and exclusion conditions, $E[\varepsilon_{it}^m \mathbf{z}_{it}] \neq 0$ & $E[\varepsilon_{it}^{-m} \mathbf{z}_{it}] = 0$ respectively. With these instruments, I can estimate equation (5) using two-stage least squares, recover column $\mathbf{b}^m$ and finally compute the dynamic response of all the local economic variables in $\mathbf{Y}_{it}$ to a structural population inflow shock ε_{it}^m .

Local Projections Estimating impulse response functions with a VAR imposes parametric restrictions on dynamics, in the sense that the joint evolution of the variables must be well approximated by a finite-lag linear system. To assess whether the estimated responses are driven by this finite-lag linear approximation VAR restriction, I also estimate impulse

³From (2) we know $u_{it}^j = Y_{it}^j - \sum_{l=1}^p \boldsymbol{\delta}_l^j \cdot \mathbf{Y}_{it-l} - \nu_i^j - \nu_t^j$. Use this in (4) and define $\boldsymbol{\Gamma}_l^j = \boldsymbol{\delta}_l^j - b_j^m \boldsymbol{\delta}_l^m$ to obtain (5).

responses to a migration inflow shock using local projections,

$$y_{i,t+h} - y_{i,t-1} = \beta_{0,h}^y + \beta_h^y \Delta m_{it} + \Gamma'_{y,h} \mathbf{X}_{it} + u_{h,it}^y.$$

Here, $y_{i,t+h}$ is the outcome of interest in commuting zone i h -periods ahead, m_{it} is the inflow rate in logs, $\mathbf{X}_{it}$ is a vector of controls that includes lags of the instrument, time and commuting-zone fixed effects, the sum of the shares, and the lagged VAR variables. In Section 3.1, I show that the estimated coefficients β_h^y are comparable to the impulse responses obtained from the VAR.

2.2 Shift-Share Instruments

To identify the effect of structural market size shocks using the methodology above, I construct a set of [Altonji and Card \(1991\)](#)-style instruments z_{it}^g , based on migration shares s_{int} and demographic shocks g_{nt} :

$$z_{it}^g = \sum_n s_{int} \cdot g_{nt} \quad (6)$$

where n is the county from which migrants originate. To see how this instrument captures variation in market size, note that we can decompose population growth $\Delta \ln N_{it}$ as:

$$\Delta \ln N_{it} \approx \frac{\Delta N_{it-1}}{N_{it-1}} + \frac{\Delta b_{it}^{net}}{N_{it-1}} + \underbrace{\sum_n \left(\frac{\tilde{m}_{int-1}}{N_{it-1}} \cdot \frac{\Delta \tilde{m}_{int}}{\tilde{m}_{int-1}} \right)}_{\text{In-migration rate } \Delta m_{it}} - \frac{\Delta \tilde{o}_{it}}{N_{it-1}} \quad (7)$$

where b_{it}^{net} is births net of deaths, $\tilde{m}_{int}$ is the total flow of migrants coming from n to i and $\tilde{o}_{it}$ is the total outflow of individuals from i .

We can see that the third term in equation (7) is the in-migration rate, which is decomposed into inflow shares and growth rates. I use the former to construct the instrument so that $s_{int} = \tilde{m}_{int-1}/N_{it-1}$. This is the “pull” factor of migration: since individuals tend to move to places that previously received migrants coming from the same area, the share s_{int} “pulls” households from n to i . On the other hand, the “push” factor will be the demographic shocks that produce an outflow from n and proxy for the growth rates within the sum in equation (7). To this end, and following a similar strategy to [Karahan, Pugsley and Şahin \(2019\)](#) and [Shimer \(2001\)](#), I set g_{nt} equal to the 25-year lags of birth rates in county n .

Note that, unlike previous migration studies that use this type of instrument, the shares are lagged by one year instead of multiple periods. Although this produces a stronger instrument it also introduces endogeneity since the predictable element of current productivity could influence past migration shares. Nevertheless, I follow [Borusyak, Hull and Jaravel](#)

(2022) and estimate the IV regression at the *shock-level*, which in this case is at the county of origin level. Thus for each dependent variable Y_{it}^j in the VAR, instead of estimating equation (5), I use the following specification:

$$\bar{Y}_{nt}^{j\perp} = b_j^m \cdot \bar{m}_{nt}^\perp + \bar{e}_{nt}^{j\perp} \quad (8)$$

where, for every variable x , we have $x^\perp$ is the residual from a regression of x_{it} on the original controls, including the lagged values of Y_{it}^j , and $\bar{x}_{nt}^\perp = \sum_i \frac{s_{int}}{s_{nt}} x_{it}^\perp$.

In addition to recovering the same coefficient of interest b_m^j , this specification has other benefits. On the one hand, it recovers the appropriate standard errors and therefore avoids the issues with classic shift-share designs shown in [Adao, Kolesár and Morales \(2019\)](#), such as the potential correlation between residuals and instruments across observations. Furthermore, this method relies on the exogeneity of the shock g_{nt} and not that of s_{int} , which is why I can use migration shares from the previous period to strengthen the instrument.

To satisfy this exclusion restriction, birth rates 25 years ago in counties sending migrants need to be exogenous to unobservable variables affecting the current dependent variables in destinations. Specifically, following [Borusyak, Hull and Jaravel \(2022\)](#), $\bar{e}_{nt}^j$ is the average unobserved shock to variable j (for example wages) across CZs that are mostly receiving migrants from n . Lagged birth rates need to be uncorrelated with these unobservable shocks. This would be violated if, for example, individuals making fertility decisions could predict economic conditions 25 years in the future in the counties receiving the most migrants after that time interval. Aggregate shocks, such as booms and recessions, as well as permanent productivity differences, are controlled for by time fixed effects and using differenced variables respectively.

2.3 Data

The IRS provides annual Statistics of Income (SOI), which include county-to-county population flows from 1990 to 2018. Specifically, for any origin i and destination j in year t , we observe the number of individual income tax returns that changed addresses from i in $t-1$ to j in t . This dataset also includes the number of exemptions as well as the total adjusted gross income associated with those returns. Following IRS guidance, the first variable proxies for the number of households that move from one county to another whereas the second represents the number of individuals that do so. Consequently, I define migration inflows m_{it} as the total exemption inflow from counties within the US. I also define outflows o_{it} in a similar manner.

The BEA also provides GDP, wages, and employment at the county level, which are some of the variables in $\mathbf{Y}_{it}$ ⁴. This data is available from 2001 to 2019. Furthermore, I obtain

⁴GDP figures correspond to CAGDP2 whereas the remaining BEA variables come from the Economic

the number of establishments for each county from the Business Dynamics Statistics (BDS) produced by the Census Bureau. To aggregate all these measures at the CZ level, I use the USDA ERS definitions from 2000, which map each county to one of 709 CZs. As described in [Autor and Dorn \(2013\)](#), these geographic areas group counties with strong commuting ties, and therefore constitute a better unit of analysis for studying the local effects of migration. Furthermore, unlike metropolitan areas that only cover certain locations, CZs encompass the entire continental US. Finally, to find housing price growth Δhp_{it} , I use the FHFA house price index, which is available at the county level. Therefore, I use the same delineation of CZs to compute an average housing index growth rate, weighted by the population of each county within a commuting zone. Table 1 shows the descriptive statistics for these variables.

TABLE 1 DESCRIPTIVE STATISTICS

	Mean	St. Dev	10th percentile	50th percentile	90th percentile	95th percentile	99th percentile
VAR variables							
Δm_{it} (%)	-1.61	32.05	-38.37	-1.06	36.01	47.97	78.45
Δo_{it} (%)	-1.62	29.20	-36.31	-0.05	32.30	43.09	68.81
Δy_{it} (%)	1.53	5.75	-3.28	1.30	6.76	10.02	19.06
Δw_{it} (%)	0.92	1.90	-0.96	0.87	2.87	3.72	6.39
Δe_{it} (%)	-0.04	2.25	-2.45	0.26	1.98	2.67	4.60
Δk_{it} (%)	-2.21	14.87	-19.26	-1.88	14.12	21.47	39.60
Δhp_{it} (%)	2.37	5.08	-3.01	2.36	7.89	10.34	16.94
Size							
Population (000s)	470.48	1,173.89	26.06	137.24	1,082.28	2,072.62	5,123.30
Employment (000s)	216.68	551.33	10.38	53.95	482.39	992.66	2,500.99
Real GDP (000s)	24,170.51	73,416.45	1,057.24	5,051.49	48,953.54	106,580.28	340,762.29
Density (Pop. per sqmi)	122.49	348.11	6.88	49.59	259.33	440.98	1144.87
Migration							
Net Inflow Rate (%)	-0.03	1.27	-0.50	-0.07	0.48	0.83	1.75
Gross Inflow Rate (%)	1.61	1.25	0.60	1.37	2.77	3.46	5.15
Gross Outflow Rate (%)	1.64	1.39	0.75	1.48	2.56	3.10	4.84
Productivity & Dynamism							
Real GDP per Worker (000s)	98.85	36.02	75.44	91.68	124.06	144.84	232.01
Real Average Wage (000s)	37.56	6.89	30.70	36.15	45.86	49.87	61.62
Employment to Population	0.42	0.07	0.33	0.42	0.52	0.54	0.60
Establishment Entry Rate	9.14	2.23	6.63	8.82	12.04	13.25	15.76

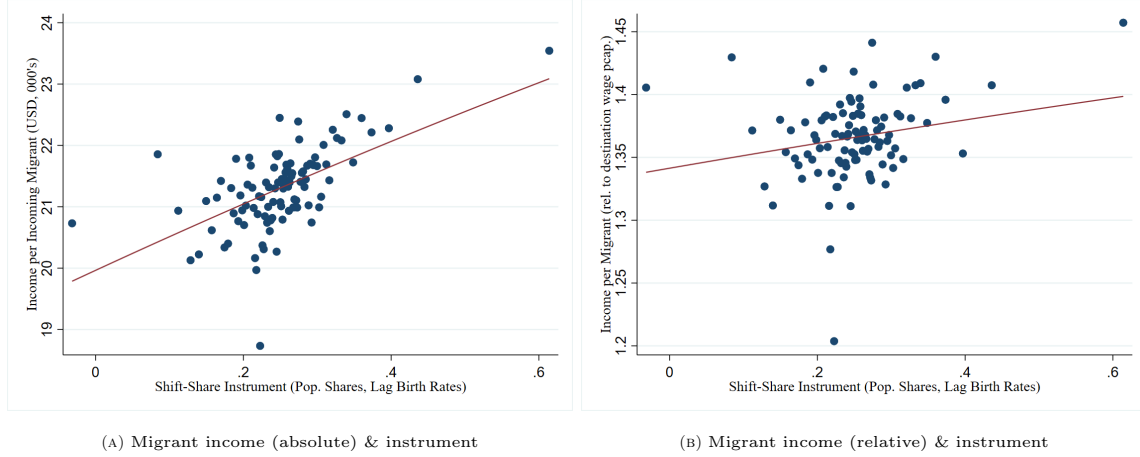
Note: Sample consists of 660 commuting zones within the continental US (excluding Alaska) and years 2002-2018.

To construct the instrument I need birth rates at the county level. I obtain this from the CDC’s National Vital Statistics microdata, which provides details on each birth during the calendar year⁵. Once I build the instrument I observe how it correlates with migrant income. The left panel of Figure 1 shows how the instrument is positively correlated with the absolute income per capita among households arriving to CZs. In other words, this suggest the “push” and “pull” factors are associated with high income migration. However, when we control for the average wage at the destination this relationship becomes flatter, as evidenced by the right panel of the same figure.

Profile for counties (CAINC30)

⁵This data is publicly available for all counties until 1988. From 1989 onwards, I only have the number of births for counties with a population over 100,000 (which corresponds to 500 counties). Consequently, I impute the missing births using the average share (out of all state births) for each county for previous years.

FIGURE 1. MIGRANT INCOME & SHIFT-SHARE INSTRUMENT



Note: Each figure is a binned scatter plot with 100 bins. Controls for year and commuting zone fixed effects.

3 The Local Effects of Migration Inflows

The first subsection estimates the SVAR-IV model described above on the entire sample. The result is a set of structural impulse responses that show the evolution of local economies after a 1% increase in the migration rate. The remaining parts of this section delve deeper into these effects by studying the heterogeneity across different types of commuting zones as well as the impact on wages in several service industries.

3.1 Baseline Impulse Responses

The shift-share structure of the instrument allows any type of shock at the origin commuting zone level g_{nt} that is exogenous with current economic conditions. Consequently, I compare lagged birth rates with other demographic “push” factors in Table 2. The first is the total outflow from CZ n : note from equation (7), this effectively replaces the specific migration flow from n to i with the total number of people leaving n . This explains why this origin-level shock produces a stronger instrument than the other two, with an F-statistic of 73.96⁶ when combined with the migration-to-population shares found in equation (7). Nevertheless, this variable is the least likely to be uncorrelated with current unobservable economic shocks affecting destinations.

Another CZ level “push” variable we can use is the total population growth rate in origin n . As column (4) in Table 2 shows, the resulting instrument is highly correlated with in-migration growth, with the highest coefficient among all three shocks. The increasing population in origin counties produces a higher outflow that gets allocated according to

⁶All the first stage results, including the F-statistic come from the *shock-level* regression developed by Borusyak, Hull and Jaravel (2022).

TABLE 2 SHOCK-LEVEL IV FIRST STAGE

	Δm_{it} (1)	Δm_{it} (2)	Δm_{it} (3)	Δm_{it} (4)	Δm_{it} (5)	Δm_{it} (6)
z_{it}^{outflow}	80.57*** (9.37)	-38.83*** (73.27)				
$z_{it}^{\text{pop. growth}}$			184.24*** (73.48)	6398.21* (3158.09)		
$z_{it}^{\text{birth rate (25y ago)}}$					0.53* (0.44)	76.78*** (16.09)
# of destination-years	8,232	8,232	8,232	8,232	8,232	8,232
# of origin-years	11242.00	11242.00	11358.00	11358.00	11358.00	11358.00
F-stat	73.96	0.28	6.29	4.10	1.47	22.79
Shares relative to total migration	✓		✓		✓	
Shares relative to total population		✓		✓		✓

Note: Robust standard errors clustered at the originating commuting zone level in parentheses; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$; All regressions control for the lagged values of the variables in $\mathbf{Y}_{it}$ and contain time fixed effects. Year range is 2002-2018. For each shock (total outflow, population growth and lagged birth rate), this table reports two sets of first stage results as a robustness check. The first uses an instrument based on the share of migration from n to i relative to total migration $\tilde{m}_{int-1}/m_{it-1}$. The second is the migration-to-population share $\tilde{m}_{int-1}/N_{it-1}$.

the migration shares. Nevertheless, its F-statistic of approximately 4.10 makes this a weak instrument. Furthermore, although total population growth is more likely to satisfy the exclusion restriction than total outflow, it can still suffer from endogeneity as it is not a lagged variable.

Finally, Table 2 also reports the first stage for the instrument that uses lagged birth rates, which is the one I use to compute the baseline IRFs later in this paper. As column (6) shows, as z_{it}^{birth} increases so does the growth rate of migration inflows. Furthermore, this coefficient is statistically significant and possesses an F-statistic of 22.79. These results, along with the arguments provided in the previous section, make this third instrument the preferred variable for estimating the response to a structural migration shock.

TABLE 3 SHOCK-LEVEL IV SECOND STAGE

	Δo_{it}		Δy_{it}		Δw_{it}		Δl_{it}		Δk_{it}		Δhp_{it}	
	OLS (1)	IV (2)	OLS (3)	IV (4)	OLS (5)	IV (6)	OLS (7)	IV (8)	OLS (9)	IV (10)	OLS (11)	IV (12)
Δm_{it}	0.12*** (0.04)	0.10 (0.14)	0.02 (0.01)	0.14* (0.05)	0.01*** (0.00)	0.05** (0.02)	0.02*** (0.00)	0.09*** (0.02)	0.01 (0.00)	0.03** (0.02)	0.02** (0.00)	0.21*** (0.10)
# of destination-years	8232.00	8,232	8232.00	8,232	8232.00	8,232	8232.00	8,232	8232.00	8,232	8232.00	8,232
# of origin-years		11358.00		11358.00		11358.00		11358.00		11358.00		11358.00

Note: Robust standard errors clustered at the originating commuting zone level in parentheses; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$; All regressions control for the lagged values of the variables in $\mathbf{Y}_{it}$ and contain time fixed effects. Year range is 2002-2018.

Table 3 reports the shock-level two-stage least squares estimates using the birth-rate instruments. Recall that these coefficients correspond to the first column of the matrix $\mathbf{B}$

in equation (2) and therefore identify the response on impact of each outcome to a migration inflow shock ε_{it}^m . The first result is that outflows rise on impact when inflows increase: the IV coefficient in column (2) is 0.10, but not statistically significant. In contrast, the OLS estimate in column (1) is 0.12 and statistically significant. This gap is consistent with upward bias in OLS if unobserved shocks are positively correlated with both inflows and outflows, such as destination-specific changes that simultaneously attract new migrants and induce churn among incumbent residents.

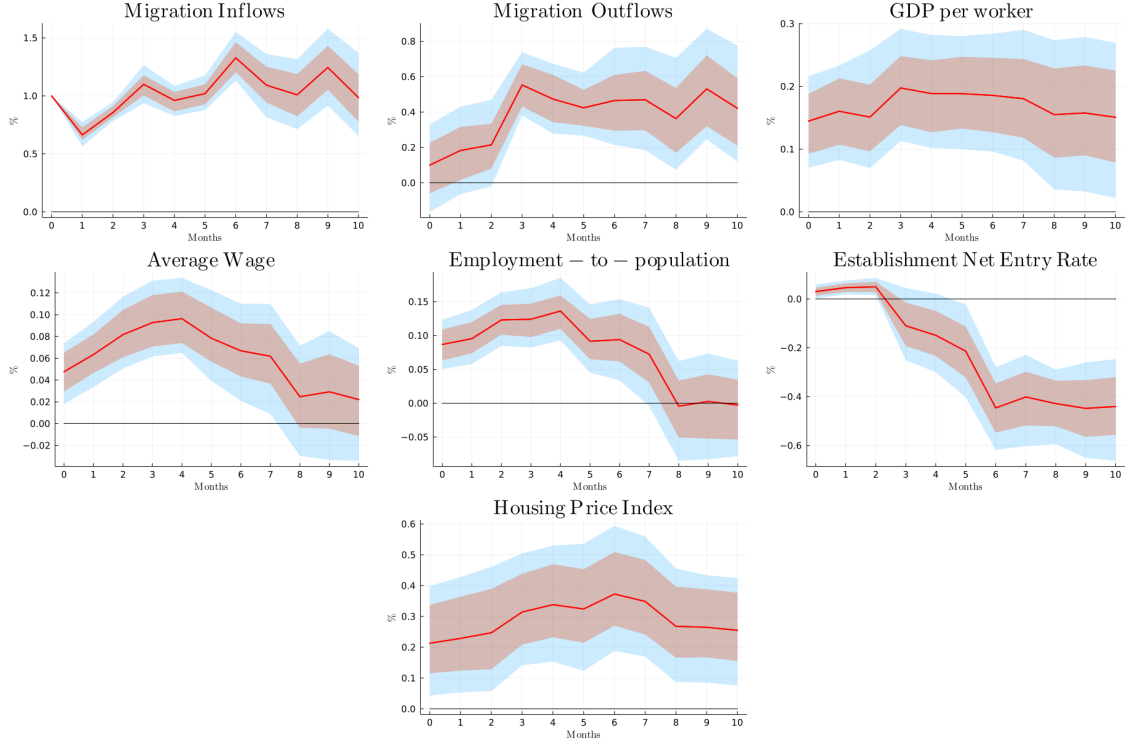
On the other hand, the contemporaneous responses of output per worker and wages are positive and statistically significant under IV, and larger than their OLS counterparts. Column (4) shows that GDP per worker rises by 0.14% on impact, while column (6) shows that average wages increase by 0.05%. These magnitudes suggest that, at least initially, the migration inflow shock acts like a positive demand shock that propagates from goods demand to labor demand. Column (8) also shows a sizable and precisely estimated rise in employment-to-population of 0.09%. Unobserved shocks that negatively affect native-dominated industries, relative to smaller ones attracting incoming migrants, could be driving the downward bias in the OLS measures. Together, these contemporaneous effects are consistent with short-run agglomeration gains, such as improved matching in thicker labor markets, operating alongside higher local demand.

Finally, column (10) indicates that net entry increases modestly on impact by 0.03. Housing prices exhibit the largest contemporaneous response among the outcomes reported: in column (12), housing prices rise by 0.21% on impact. The size of the housing response is suggestive of congestion forces: when inflows raise local housing demand, prices can respond quickly in places where short-run housing supply is relatively inelastic. This immediate increase in housing costs is a natural candidate to counteract the demand-driven gains in wages and productivity as the adjustment unfolds. To evaluate these dynamics, I now turn to the structural impulse responses.

After a 1% increase in the migration inflow rate, the inflow response is large and persistent over the horizons we observe. Figure 2 decomposes the dynamics into inflows and outflows. Inflows jump by 1 on impact and remain close to that level through year 10, with some non-monotonic adjustment: they decrease in years 1–2 and then rebound, peaking at 1.33% in year 6. After their initial increase on impact, outflows also rise following the inflow shock and reach roughly 0.55% by year 3, and remain elevated thereafter. Importantly, the increase in outflows does not indicate a deterioration of local labor-market outcomes—indeed, wages and productivity rise—but is consistent with displacement/sorting forces: as population rises, housing and other congestion-related costs increase, pushing some (marginal) residents to leave even as the CZ becomes more productive.

In the years following the inflow shock, local market size expands and economic activity responds. From the second stage results we saw output per worker increases immediately.

FIGURE 2. CUMULATIVE IRFS FOR ECONOMIC VARIABLES



Note: The red line represents the cumulative impulse response. The blue shade indicates a 90% confidence level. Standard errors are generated by Monte-Carlo with 200 repetitions.

Figure 2 shows it then reaches its peak around year 3, remaining positive through year 10 at 0.15%. Average wages also rise on impact and display a clear hump shape, peaking in year 4 at 0.09% although the magnitude of the effect is lower than that of average productivity. Employment-to-population increases in the short run, peaking at 0.14% by year 4, but then reverts back toward zero and is essentially flat in the longer term. These patterns are consistent with short-run agglomeration forces: a thicker labor market and stronger local demand improve matching and utilization and can raise measured productivity, as in [Duranton and Puga \(2004\)](#). At the same time, the fact that the employment-to-population response fades suggests that population is increasing faster than employment in the medium run (e.g., due to composition or participation), rather than a sustained boom in the employment share.

Nevertheless, housing prices also increase strongly and persistently, exerting pressure on residents' budgets. Figure 2 shows how the response of housing prices rises from 0.21% on impact to a peak of 0.37%, and remains elevated at 0.26% by year 10. Relative to output per worker and wages, which peak at 0.2% and 0.09%, the housing-price response is larger, consistent with congestion forces that tighten local land/housing markets. Moreover, higher housing values likely proxy for a broader set of population-linked externalities, such as commuting congestion, pollution, permitting delays and higher commercial rents. These

congestion costs help rationalize why outflows rise alongside inflows: some agents are priced out or choose to re-optimize away from the CZ even though nominal wages and productivity increase.

These congestion forces also show up clearly on the firm extensive margin. Although net entry experiences a small increase on impact, it turns sharply negative after year 3, reaching -0.45% at year 6 and remains around that level for the following years. This, along with the housing price response, suggests rising rents and other congestion-related fixed costs tax entry and the viability of marginal firms. In other words, agglomeration gains appear to accrue primarily to incumbents, which are typically larger more productive firms that can expand on the intensive margin, while higher land/operating costs discourage new firm formation and reduce the number of establishments. In the additional results of Section 3.2 we will see this is consistent with other measures of establishment and job dynamism.

Local Projections As we saw in Section 2.1, estimating a VAR assumes we can model the joint dynamics of all variables with a finite-order lag system. Consequently, to test whether the resulting impulse responses rely on this assumption I re-estimate them using local projections. Figure 3 shows the results. Most of the variables of interest, including wages and output per worker, behave similarly after a migration inflow shock across both methods.

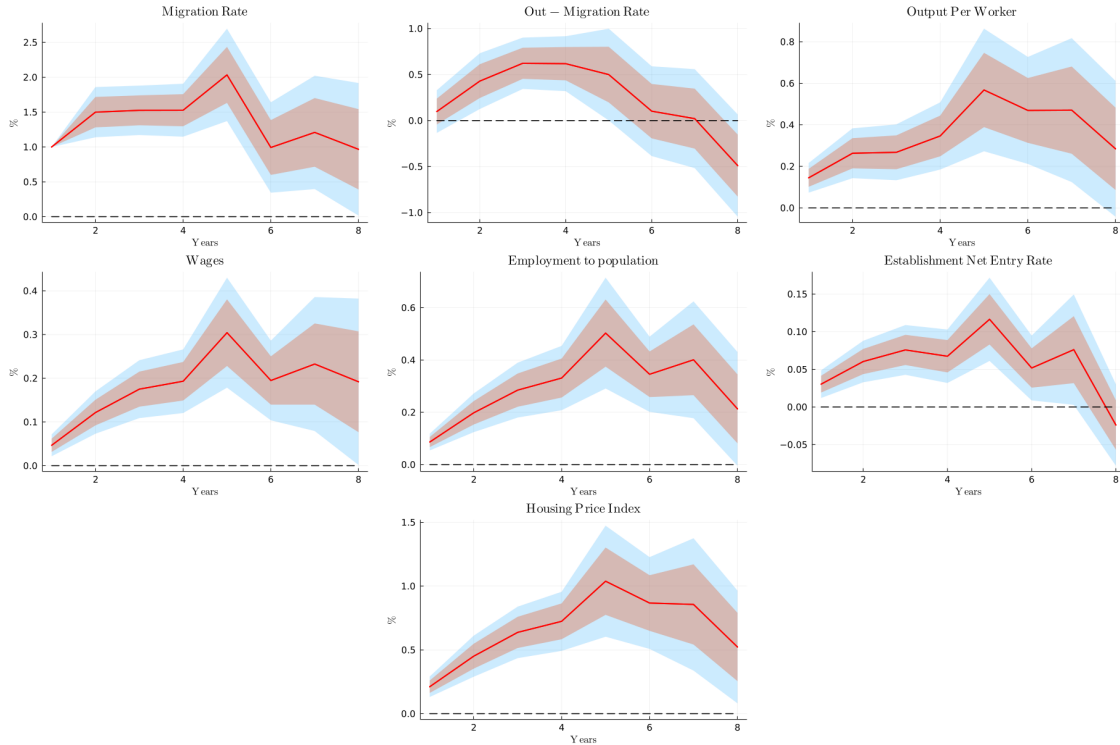
The only variable that exhibits a different dynamic response is the establishment net-entry rate. Whereas Figure 2 shows a decline after two years, the local-projection estimates imply an increase in the entry rate over the same horizons. Although both methods deliver a similar impact effect, the VAR-based responses are obtained by iterating an estimated finite-lag linear system; as a result, small lag-length misspecification or estimation error in the entry equation can accumulate and generate a different propagation path at longer horizons. By contrast, LP-IV estimates each horizon directly and is therefore less sensitive to VAR recursion, so the divergence is more consistent with differences in how the two approaches discipline the dynamics of entry than with differences in shock identification.

3.2 Additional Dynamics and Heterogeneous Effects

As inflows rise, a commuting zone sees persistent increases in output per worker and housing prices, while employment and wages respond more temporarily. To better understand the forces behind these patterns, I next examine how inflows affect other measures of local dynamism and how the responses vary across sectors and across different types of commuting zones.

Establishment Dynamics and Job Creation In the baseline results, the establishment net entry rate decreases after a positive inflow shock. I now augment the VAR by adding

FIGURE 3. LOCAL PROJECTION CUMULATIVE IRFs



Note: The red line represents the cumulative impulse response. The blue shade indicates a 90% confidence level. Standard errors are generated by Monte-Carlo with 200 repetitions.

different measures of dynamism one at a time.⁷, starting with the gross entry rate. The drop we observe in Figure 4 reveals that most of the decrease in the net rate comes from a drop in the number of firms entering as the gross entry rate reaches a low of -0.2% after five years. Furthermore, establishments per worker decline throughout, reaching a level between -0.13% and -0.16% after three years. Finally, the job-flow responses reinforce the same narrative. Job destruction drops sharply and persistently to roughly -0.3% after four years, indicating more stable ongoing matches and fewer separations after the inflow shock. In contrast, the response of job creation is comparatively small and not statistically significant.

Taken together, these outcomes point to an adjustment in which incumbent firms become larger and stabilize employment relationships through potential agglomeration and matching benefits. On the other hand, housing prices, which proxy for congestion externalities, negatively affects the extensive margin through net entry and elevated outflows. This combination helps explain why outcomes such as productivity remain positive, while housing prices stay elevated and firm entry remains depressed: the local economy moves toward a medium-run balance between agglomeration benefits and congestion costs.

⁷To estimate each of the impulse responses Figure 4, I estimate a VAR with the original 7 variables, along with the variable in the figure.

FIGURE 4. CUMULATIVE IRFS FOR ESTABLISHMENTS AND JOBS



Note: The red line represents the cumulative impulse response. The blue shade indicates a 90% confidence level. Standard errors are generated by Monte-Carlo with 200 repetitions.

Wage Effect by Sector Recall from Figure 1 that migrant income is positively correlated with the instrument. Because the identifying variation is driven by relatively high-income migrants, part of the baseline wage response in Figure 2 could reflect composition. To assess whether wage effects are broader than composition, I examine wage responses within industries that employ relatively homogeneous labor in terms of skill. If wages move within these sectors, that suggests in-migration affects local pay more generally rather than only changing who lives in the commuting zone. To do so, I select a set of 3-digit NAICS industries and re-estimate the VAR multiple times, each time adding the average wage for one industry as an additional outcome, analogous to the exercise with job dynamics.⁸ Figure 5 reports the resulting impulse responses.

The results suggest the baseline wage increase is not purely a composition effect. For reference, the aggregate wage response in Figure 2 rises modestly in the short run and peaks early between 0.05–0.10% over the first few years in the baseline specification. In Figure 5, wages also rise within several lower-skill, locally oriented industries: trucking/transportation and food services, for example, exhibit peak increases on the order of 0.2–0.3%. These

⁸The Quarterly Census of Employment and Wages (QCEW) provides average wages at the county-industry level. To aggregate them at the CZ level, I use the same delineation detailed in Section 2. Each industry is therefore associated with its own VAR specification.

patterns are consistent with an expansion in local market size that raises demand for non-tradable services and strengthens matching in thick labor markets, generating short-run agglomeration gains that translate into higher pay for some workers.

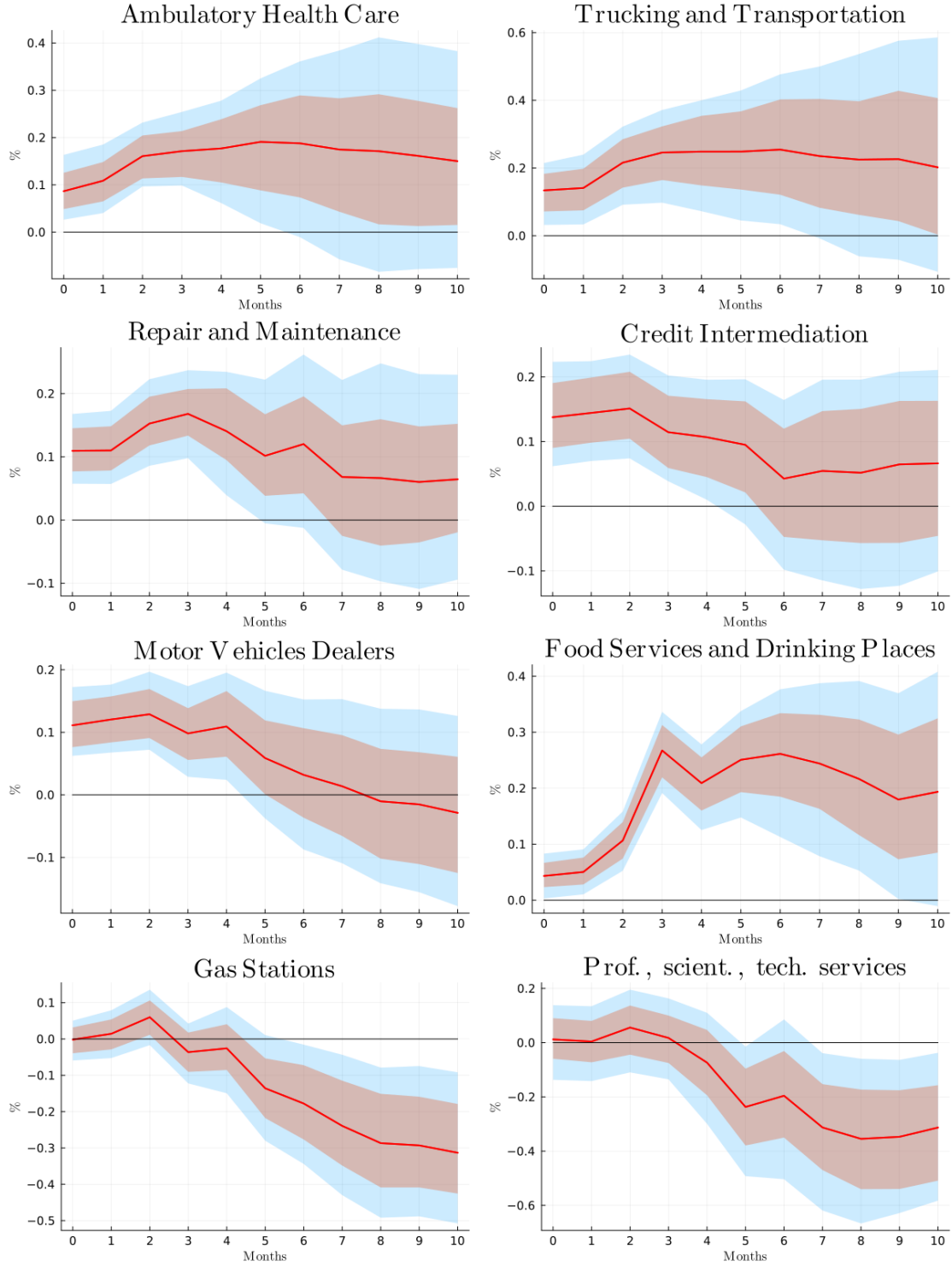
At the same time, other sectors display weaker or even negative wage responses at longer horizons. Industries such as gas stations and professional and technical services, for instance, show wage declines of roughly -0.3% several years after the shock. This heterogeneity fits the broader narrative in which agglomeration forces coexist with congestion: inflows raise housing and operating costs, which can compress margins in some activities and shift local adjustment toward incumbent expansion and sectoral reallocation rather than broad-based wage growth. In other words, incoming migrants appear to increase wages in some industries through market-size and agglomeration channels, even as congestion and cost pressures produce uneven longer-run effects across sectors.

Responses by Commuting Zones Density and Population The previous section highlighted that local responses to migration inflows can vary over time. I now study heterogeneity across commuting zones (CZs), focusing on housing prices. The reason different densities could produce heterogeneous responses is the differential relevance of congestion and agglomeration forces. Workers in New York City interact with each other more frequently than those in Phoenix, where land is less densely populated, which facilitates the human capital externalities that [Lucas \(1988\)](#) described. Nevertheless, these areas also experience higher congestion, and therefore the net effect of migration shocks is ambiguous. To construct the groups, I compute each CZ’s average population and population density over the sample period and classify CZs into three groups: low, medium and high. I then re-estimate the SVAR-IV model separately for each group.

In [Figure 6](#), I report the resulting impulse responses for housing prices by CZ group. This is the only outcome that displays substantial cross-group variation in the magnitude of its response. Intuitively, housing prices are a natural proxy for congestion costs: when inflows raise local housing demand, price pressure should be stronger where short-run housing supply is tighter and weaker where supply can adjust more easily. Because supply tightness is closely related to density and market size, the housing-price response is therefore expected to vary systematically across these groups.

[Figure 6](#) shows that housing prices rise across all CZ types, but the magnitude and timing differ markedly by density and population. By density, high-density CZs exhibit a large and rapid increase: housing prices rise by about 0.33% on impact and peak around 0.63% by year three, remaining positive at roughly 0.36% by year ten. Medium-density CZs display a smaller and smoother response, increasing by about 0.23% on impact and only peaking near 0.34% . Low-density CZs, on the other hand, show a more delayed adjustment reaching their peak of 0.61% later in year six. A similar pattern emerges by population size. High-population CZs experience the strongest and earliest appreciation, with a peak of 0.7%

FIGURE 5. CIRFs FOR WAGES ACROSS NAICS SECTORS



Note: The red line represents the cumulative impulse response. The blue shade indicates a 90% confidence level. Standard errors are generated by Monte-Carlo with 200 repetitions.

FIGURE 6. HOUSING PRICE RESPONSE BY CZ DENSITY AND POPULATION



Note: The red line represents the cumulative impulse response. The blue shade indicates a 90% confidence level. Standard errors are generated by Monte-Carlo with 200 repetitions.

by year three, maintaining an elevated level of 0.46% by year ten. In contrast, CZs with a low-population exhibit the smallest and least persistent response: prices peak around 0.35% in year six, and are close to zero by year ten. Overall, these patterns are consistent with the congestion mechanism emphasized above: migration inflows bid up housing demand everywhere, but the price response is largest and most persistent in larger, denser markets where supply is less elastic and congestion pressures are more acute, potentially feeding back into displacement and sorting.

4 Theoretical Model

Increasing migration inflows by 1% produces a positive and sustained change in local economic outcomes such as wages, output and housing prices. However, the shape of the effect across time suggests the interaction between agglomeration and congestion externalities results in different effects at different intervals. To rationalize these results, this section develops a theoretical model that will later be disciplined with the structural impulse responses.

4.1 Environment

Households The economy consists of $g \in G$ regions in continuous time $t \in \mathbb{R}^+$ and a mass $N(t)$ of workers who discount the future at rate ρ and exit the labor force at regional

rate δ . Furthermore, each period, workers enter the regional markets at an exogenous rate $b_g(t)$ are born. When they are born workers are endowed with a permanent productivity level $z \in \Omega_z$, which they draw from an initial distribution $z \sim F(z)$. They derive utility from consumption $C_g(t)$ and housing $H_g(t)$, as well as an amenity $D_g(t)$ so that:

$$U_g(C_g(t), H_g(t)) = D_g(t) \cdot C_g(t)^\alpha \cdot H_g(t)^{1-\alpha}$$

where $D_g(t) = D_g \cdot N_g(t)^{-\phi}$ and $\phi > 0$. Note that amenities are decreasing with local population. This is to account for congestion externalities other than housing such traffic or the public goods. The consumption good is a CES aggregate from differentiated regional varieties $C_g = (\sum_r c_{rg}^{\frac{\sigma-1}{\sigma}})^{\frac{\sigma}{\sigma-1}}$. To buy variety r in g , households need to pay $p_{rg}(t)$, whereas $P_g^h(t)$ is the price for each unit of housing they consume.

Each period thereafter they receive a migration opportunity, whose arrival follows a Poisson process with rate λ ⁹. This parameter captures the fixed costs associated with moving to new markets. This is one of the two mobility frictions the model includes. After receiving the migration opportunity, workers draw a preference shock for each location $\xi_g(t)$, which are distributed i.i.d. Frechet with shape parameter ν . The second friction is a flow-utility cost κ_{gd} that the household pays for moving from g to d , where $\kappa_{gd} < 1$ for $g \neq d$ and $\kappa_{gg} = 1$. Thus workers consider the value of living in each location, given these mobility costs and preference shocks that are both multiplicative and permanent as in [Crews \(2023\)](#).

The mobility decisions of workers, along with their entry and exit from the economy, will determine the spatial distribution of z at time t denoted by $\psi_g(z, t)$. Thus the total labor available in region g will be given by

$$L_g(t) = N(t) \cdot \int_{z \in \Omega_z} z \cdot \psi_g(z, t) dz$$

since workers supply their efficiency units z inelastically in the region g they choose to live in, in exchange for the local wage $W_g(t)$. Additionally, the population of region g is determined by $N_g(t) = N(t) \cdot \int_{z \in \Omega_z} \psi_g(z, t) dz$.

Production Perfectly competitive local producers supply housing services $H_g(t)$ by combining land $T_g(t)$ and labor $L_g^h(t)$ as inputs using the following function:

$$H_g(t) = \Gamma_h T_g(t)^\theta (L_g^h(t))^{1-\theta}.$$

where $\Gamma_h = \theta^{-\theta} (1 - \theta)^{-(1-\theta)}$. I assume land is a fixed factor owned by immobile landlords who collect rents $R_g(t)$.

⁹I assume that newborns do not receive this migration opportunity and are not subject to the exit probability.

Following [Romer \(1990\)](#), production of regional varieties is subject to variety gains. Each period, under perfect competition, a local firm sources $y_{gd}(j, t)$ at price $q_g(j, t)$ from firm j to produce y_{gd} , the amount of variety $g \in G$ it ships to destination d . These inputs are aggregated following:

$$y_{gd}(t) = \frac{1}{\tau_{gd}} \left(\int_0^{M_g(t)} y_{gd}(j, t)^{\frac{\varepsilon-1}{\varepsilon}} dj \right)^{\frac{\varepsilon}{\varepsilon-1}}$$

where $M_g(t)$ is the mass of intermediate input producers in g at time t , and τ_{gd} represents the usual iceberg trade cost. Local intermediate producers on the other hand face monopolistic competition as well as labor requirements given by

$$l_g(j, t) = \frac{y_g(j, t)}{A_g} + F_g(t)$$

where A_g is the fixed productivity in region g and $F_g(t)$ is the fixed cost in terms of labor units that firms have to pay. Similar to the discrete time setting in [Peters \(2022\)](#), this related to the growth rate of the mass of firms as well as the *inter-temporal knowledge elasticity* η :

$$F_g(t) = \gamma_M(t) \cdot M_g(t)^{-\eta} \quad (9)$$

where $\gamma_M(t) = \dot{M}_g(t)/M_g(t)$.

4.2 Static Equilibrium

At any point in time t , the state variables for each geography $g \in G$ consist of the mass of producers $M_g(t)$ and the distribution of efficiency units $\psi_g(z, t)$. We can therefore define a static equilibrium as follows:

Definition 1. For each $g \in G$, given $M_g(t)$ and $\psi_g(z, t)$, a **static equilibrium** in period $t \in \mathbb{R}_+$ is a set of prices and allocations for skill $\{W_g(t), L_g(t)\}$, consumption imported from all regions $\{c_{rg}(t), p_{rg}(t)\}_{r \in G}$, intermediate inputs $\{q_g(j, t), y_{gd}(j, t)\}_{j \in M_g(t), d \in G}$, housing $\{H_g(t), P_g^h(t)\}$ as well as land $\{T_g(t), R_g(t)\}$ such that given prices i) households and firms behave optimally, ii) allocations clear their respective markets.

Given their place of residence, households decide how much to consume of each region's variety with a local price index given by $P_g(t) = (\sum_r p_{rg}(t)^{1-\sigma})^{1-\sigma}$. Worker z will spend a constant share of her income $v_g(z, t) = z \cdot W_g(t)$ on consumption goods and housing services. Since land is fixed the supply of housing services under perfect competition will be:

$$H_g(t) = \left(\frac{P_g^h(t)}{W_g(t)} \right)^{\frac{1-\theta}{\theta}} \cdot \frac{T_g}{\theta} \quad (10)$$

On the other hand, local producers decide how much of their output they allocate to each region (including their own) as well as the demand for local varieties, which will be

$$y_g(j, t) = \left(\frac{Q_g(t)}{q_g(j, t)} \right)^\varepsilon Y_g(t),$$

where $Y_g(t)$ is the total output of region g and $Q_g(t)$ is the price index these firms face. Intermediate producers take this demand and choose prices and labor so as to charge a markup over marginal cost $q_g(j, t) = \frac{\varepsilon}{\varepsilon-1} \cdot \frac{W_g(t)}{A_g}$, which implies $Q_g(t) = M_g(t)^{\frac{1}{1-\varepsilon}} \left(\frac{\varepsilon}{\varepsilon-1} \cdot \frac{W_g(t)}{A_g} \right)$. Furthermore, free entry determines total labor demand for production in region g :

$$L_g^y(t) = \int_0^{M_g(t)} l_g(j, t) dj = M_g(t) F_g(t) \varepsilon \quad (11)$$

Market clearing for labor states demand from the goods and housing sector has to equal total supply $L_g(t)$. This, along with zero profits and housing market clearing, sets the total efficiency units in housing production as a constant share of regional labor: $L_g^h(t) = (1 - \theta)(1 - \alpha)L_g(t)$. Total income in region g is $W_g(t)L_g(t)$ and balanced trade implies:

$$W_g(t)L_g(t) = \sum_d \pi_{gd}(t) W_d(t) L_d(t) \quad (12)$$

where $\pi_{gd}(t)$ is the share of consumption expenditures in d spent in goods from g . This in turn is given by:

$$\pi_{gd}(t) = \frac{\left(\tau_{gd} \cdot Q_g(t) \right)^{1-\sigma}}{\sum_{i \in G} \left(\tau_{id} \cdot Q_i(t) \right)^{1-\sigma}} \quad (13)$$

Using equilibrium input prices I obtain the price index $P_g(t) = Q_g(t) \cdot \pi_{gg}(t)^{\frac{1}{\sigma-1}}$ as well as the price of housing units in region g given by:

$$P_g^h(t) = \left(\frac{(1 - \alpha) \cdot L_g(t) \cdot \theta}{T_g} \right)^\theta \cdot W_g(t) \quad (14)$$

Combining them with the demand for consumption goods and housing, I derive the flow utility a worker with productivity level z receives when living in region g :

$$U_g(z, t) = \underbrace{D_g(t)}_{\text{Amenities}} \cdot \underbrace{z \cdot \frac{\Gamma_g}{\Gamma_U} \cdot \left(\pi_{gg}(t) \right)^{\frac{1-\alpha}{1-\sigma}} \cdot \left(M_g(t) \right)^{\frac{\alpha}{\varepsilon-1}} \cdot \left(L_g(t) \right)^{-\theta(1-\alpha)}}_{\text{Real Wage}} \quad (15)$$

where $\Gamma_U = \left(\frac{\varepsilon}{\varepsilon-1} \right)^\alpha ((1 - \alpha)\theta)^{\theta(1-\alpha)}$ and $\Gamma_g = A_g^\alpha T_g^{\theta(1-\alpha)}$. Note the real wage is a function of the mass of firms and total efficiency units in the regional market, as well as those in

other localities through the own trade share $\pi_{gg}(t)$.

4.3 Dynamic Equilibrium and Balanced Growth Path

Now that we defined a static equilibrium for every t we can study the dynamics of the model, which involves spatial labor reallocation through migration as well as the evolution of the mass of firms within each region.

The following Hamilton-Jacobi-Bellman (HJB) equation summarizes the household's recursive problem (I derive this equation in Appendix A.1.1):

$$(\delta + \rho)V_g(z, t) = U_g(z, t) + \dot{V}_g + \lambda \sum_i m_{gi}(z, t) \left[\delta_{gi}(z, t) V_i(z, t) - V_g(z, t) \right] \quad (16)$$

where $\delta_{gi}(z, t) = \frac{1}{G} \cdot \kappa_{gi} \cdot m_{gi}(z, t)^{-(\frac{1}{\nu}+1)}$ and the migration shares are given by:

$$m_{gd}(z, t) = \frac{[\kappa_{gd} \cdot V_d(z, t)]^\nu}{\sum_i [\kappa_{gi} \cdot V_i(z, t)]^\nu} \quad (17)$$

This is the probability that a worker with productivity z moves from region g to d , conditional on having a migration opportunity mediated by λ .

Recall that the distribution of individual productivities $\psi_g(z, t)$ will determine regional labor and population. Thus the evolution of these variables will depend on the Kolmogorov forward equation below (see Appendix A.1.2 for a derivation):

$$\begin{aligned} \dot{\psi}_g(z, t) &= b_g(t) \cdot \frac{N_g(t)}{N(t)} \cdot f(z) - \delta \psi_g(z, t) \\ &- \lambda \left[(1 - m_{gg}(z, t)) \psi_g(z, t) - \sum_{i \neq g} m_{ig}(z, t) \psi_i(z, t) \right] \\ &- \psi_g(z, t) \frac{\dot{N}(t)}{N(t)} \end{aligned} \quad (18)$$

Finally, equation (11) establishes a relation between the fixed cost, labor demand and the mass of firms. Namely, regions with a lower $F_g(t)$ will have a smaller individual labor requirement for each producer which tends to reduce total demand $L_g^y(t)$ in that market. Although this cost is fixed at any point t , equation (9) states its evolution will depend on that of the mass of firms $M_g(t)$. By combining both I obtain:

$$\dot{M}_g(t) = \left(\frac{\alpha + \theta(1 - \alpha)}{\varepsilon} \right) L_g(t) M_g(t)^\eta \quad (19)$$

This differential equation determines how the regional mass of firms changes over time. I

can now define a dynamic equilibrium.

Definition 2. A *dynamic equilibrium* is i) a value function $V_g(z, t)$, ii) a distribution $\psi_g(z, t)$ and iii) migration shares $m_{gd}(z, t)$ for $g, d \in G$, $z \in \Omega_z$ and $t \in \mathbb{R}_+$, as well as iv) functions for labor, population and mass of firms $\{L, N, M\}$ and v) wages $\{W\}$ such that:

1. M evolves according to equation (19);
2. V and m solve the HJB equation in (16), taking $\{L, N, M\}$ and $\{W\}$ as given;
3. the densities ψ evolve according to the Kolmogorov equation (18) $\forall r, g \in G$, taking migration decisions as given;
4. given ψ , populations and total skills satisfy the following equations:

$$N_g(t) = N(t) \cdot \int_{\Omega_z} \psi_g(z, t) dz \quad L_g(t) = N(t) \cdot \int_{\Omega_z} z \cdot \psi_g(z, t) dz \quad N(t) = \sum_{g \in G} N_g(t)$$

5. the resulting prices and allocations constitute a static equilibrium $\forall t \in \mathbb{R}_+$.

Balanced Growth Path A balanced growth path (BGP) is a special type of dynamic equilibrium where all variables grow at a constant rate. To define such a solution, I detrend each variable $x(t)$ by rewriting it relative to its long-run growth rate along the BGP $x(t) = \tilde{x}(t) \cdot e^{\gamma_x t}$. For all derivations see Appendix A.1.3. The result is a set of detrended equilibrium conditions.

The first is a differential equation describing the evolution of regional varieties: the detrended version of equation (19) in terms of growth rates. That is

$$\gamma_{\tilde{M}_g}(t) + \gamma_M = \left(\frac{\alpha + \theta(1 - \alpha)}{\varepsilon} \right) \tilde{L}_g(t) \tilde{M}_g(t)^{\eta-1}. \quad (20)$$

From this equation I also find $\gamma_L = \gamma_M(1 - \eta)$, thereby connecting the growth rate of $M_g(t)$ and that of $L_g(t)$ along the BGP. I also detrend the HJB in equation (16) to obtain:

$$\begin{aligned} (\delta + \rho - \gamma_v) \tilde{V}_g(z, t) &= \tilde{U}_g(z, t) + \partial_t \tilde{V}_g(z, t) \\ &+ \lambda \sum_i m_{gi}(z, t) \left[\delta_{gi}(z, t) \tilde{V}_i(z, t) - \tilde{V}_g(z, t) \right], \end{aligned} \quad (21)$$

where the effective discount rate is now $(\delta + \rho - \gamma_v)$ with $\gamma_v = \left(\frac{\alpha}{\varepsilon - 1} - (1 - \eta)(\phi + \theta(1 - \alpha)) \right) \gamma_M$.

The detrended Kolmogorov forward equation (18) is now:

$$\begin{aligned}
\partial_t \tilde{\psi}_g(z, t) &= b_g(t) \frac{\tilde{N}_g(t)}{\tilde{N}(t)} f(z) - \delta \tilde{\psi}_g(z, t) \\
&- \lambda \left[(1 - m_{gg}(z, t)) \tilde{\psi}_g(z, t) - \sum_{i \neq g} m_{ig}(z, t) \tilde{\psi}_i(z, t) \right] \\
&- \left(\tilde{\gamma}_N(t) + \gamma_N \right) \tilde{\psi}_g(z, t).
\end{aligned} \tag{22}$$

By detrending this equation I find $\gamma_\psi = 0$, which implies from the definition of $L_g(t)$, that labor grows at the same rate as population along the BGP so that $\gamma_L = \gamma_N$. The detrended versions of these variables are defined by

$$\tilde{N}_g(t) = \tilde{N}(t) \int_{z \in \Omega_z} \tilde{\psi}_g(z, t) dz \quad \& \quad \tilde{L}_g(t) = \tilde{N}(t) \int_{z \in \Omega_z} z \tilde{\psi}_g(z, t) dz.$$

Using population dynamics, I compute the growth rate of $\tilde{N}_g(t)$ as a function of regional birth rates and δ :

$$\frac{\dot{\tilde{N}}(t)}{\tilde{N}(t)} = \frac{\sum_g b_g(t) \cdot \tilde{N}_g(t)}{\tilde{N}(t)} - \delta - \gamma_N \tag{23}$$

Note that once the overall economy reaches the BGP, all detrended variables $\tilde{x}(t)$ are constant so that $\tilde{x}(t) = \bar{x} \forall t$. I now formally define this type of equilibrium below.

Definition 3. A **balanced growth path** is a dynamic equilibrium in which population $N_g(t)$ grows at a constant rate γ_N across locations, with constant growth rates for all equilibrium variables and detrended functions $\{\tilde{V}, m, \tilde{\psi}\}$ such that:

$$\begin{aligned}
V_g(z, t) &= e^{\gamma_v t} \bar{V}_g(z) \\
m_{ig}(z, t) &= \bar{m}_{ig}(z) \\
\psi_g(z, t) &= \bar{\psi}_g(z)
\end{aligned}$$

where $\{V, m, \psi\}$ solve the dynamic equilibrium, $\{\bar{V}, \bar{m}, \bar{\psi}\}$ are the detrended value function, migration shares and skill distributions at the BGP, $\gamma_v = (\frac{\alpha}{\varepsilon-1} - (1-\eta)(\phi + \theta(1-\alpha)))\gamma_M$ and $\gamma_N = \gamma_M(1-\eta)$.

Along the BGP, the detrended HJB (21) becomes:

$$(\delta + \rho - \gamma_v) \bar{V}_g(z) = \bar{U}_g(z) + \lambda \sum_i \bar{m}_{gi}(z) \left[\bar{\delta}_{gi}(z) \bar{V}_i(z) - \bar{V}_g(z) \right] \tag{24}$$

whereas the Kolmogorov equation (22) is now

$$0 = \bar{b}_g \frac{\bar{N}_g}{\bar{N}} f(z) - (\delta + \gamma_N) \bar{\psi}_g(z) - \lambda(1 - \bar{m}_{gg}(z)) \bar{\psi}_g(z) + \lambda \sum_{i \neq g} \bar{m}_{ig}(z) \bar{\psi}_i(z) \quad (25)$$

Finally, to find the population growth rate γ_N , I use the fact that $\dot{\bar{N}} = 0$ along the BGP in equation 23. This implies:

$$\gamma_N = \frac{\sum_g \bar{b}_g}{\bar{N}} - \delta \quad (26)$$

Note that I could also derive this by integrating the BGP Kolmogorov equation (25) over all z and summing over g , as well as using the fact that along the BGP there is no net migration.

4.4 Model Solution

To compute the solution of this model, I adapt the algorithms in Achdou et al. (2022). This was done in a spatial setting in Crews (2023) to compute the steady state for a centralized economy. However, since an estimation of the structural parameters through impulse response matching requires a simulation of dynamic outcomes, I implement a fixed-point procedure that solves the transition path of the economy as well as a BGP algorithm. To simplify notation I group all general parameters into Θ and regional fundamentals into Θ_G . The full algorithms are detailed in Appendix B.

5 Structural Estimation

With the framework in the previous section I can study the effects of local migration and relocation shocks on aggregate outcomes such as output and productivity. To carry this out, I first estimate the structural parameters so that the model is consistent with the impulse responses in Section 3.

5.1 Estimation Strategy

The model parametrization includes a tuple of regional fundamentals $\{T_g, A_g, D_g\}$ as well as 14 structural parameters: those I set exogenously $\Theta_1 = \{\alpha, \rho, \sigma, \zeta_\tau, \gamma_L, \bar{N}\}$ and those I estimate within the simulation

$$\Theta_2 = \{\varepsilon, \eta, \nu, \phi, \zeta_m, \theta, \lambda, \delta\}.$$

Given a set of parameters I compute the initial and final steady states, as well as the transition path that results from an exogenous series of regional birth rates $\{b_g(t)\}$ that

take the economy away from the first BGP. This produces a simulated dataset I then use to replicate the shift-share IV and estimate the SVAR-IV described in Section 2.

Clustering Commuting Zones Solving the theoretical model is time-consuming and the cost increases dramatically with the number of regions. Therefore I need to reduce this set from the 660 CZs used in the empirical part to estimate the structural parameters of the theory. To carry this out, I applied a K-means clustering procedure based on geographical coordinates and the variables in the VAR.

In particular, for a given number of clusters, I use the dataset with all CZs in 2010 and perform a K-means clustering algorithm where the “features” are the variables in the empirical VAR as well as the CZ coordinates. This produces a set of clusters containing CZs that are similar in terms of migration, output and other economic variables, but that are also geographically close to each other. This rules out, for example, a cluster with some CZs in California as well as Florida. I then aggregate the VAR variables at the cluster level and re-estimate the SVAR-IV to compare the IRF coefficients with the ones in Section 3. I repeat these steps for a number of clusters ranging from 15 to 100, and choose the grouping that produces the coefficients closest to the ones in the baseline estimation. The result is set of 35 CZ-clusters.

Exogenous Parameters I set $\rho = 0.1$ as in [Crews \(2023\)](#) and $\sigma = 2$ following the results in [Boehm, Levchenko and Pandalai-Nayar \(2023\)](#). I target the average share of personal consumption expenditures on housing over the time period and set $\alpha = 0.89$. For the initial distribution of productivities $f(z)$, I use a truncated *Lognormal*(μ_f, σ_f) distribution using the estimates of the mean and coefficient of variation from [Huggett, Ventura and Yaron \(2006\)](#) to set μ_f and σ_f . To identify available land for housing T_g , I use raster data from the USGS National Land Cover as well as Data Elevation Models. Specifically, for each county I find the area (in square miles) that does not contain open water, wetlands or perennial snow as well as with an inclination lower than 15% following architectural guidelines and [Saiz \(2010\)](#).

I follow [Peters \(2022\)](#) parameterize bilateral trade and migration costs as functions of distance by setting $\tau_{gd} = (d_{gd}/d_{min})^{\zeta_\tau}$ and $\kappa_{gd} = (d_{gd}/d_{min})^{\zeta_m}$, with $d_{gg} = d_{min}$. Since CZ group one or more counties, I can compute the average distance between them. Therefore I use the 5% quintile of these within-CZ average distances to set $d_{min} = 32\text{mi}$. Although I estimate ζ_m through impulse response matching, I estimate ζ_τ by regressing bilateral prices on distance, similar to [Castro-Vincenzi et al. \(2024\)](#).

In particular, I use the Commodity Flow Survey (CFS) datasets from 2012 and 2017, which provide information on domestic freight shipping within the US. For each shipment i , I observe its weight and value, as well as the great circle distance $d_{i,t}$ between shipment origin and destination. Thus, since in the model $p_{gd}(t) \propto \tau_{gd}$, I estimate the following

regression:

$$\log p_{i,t} = \beta_0^p + \zeta_\tau \log(d_{i,t}) + \mathbf{X}_{i,t} + \epsilon_{i,t} \quad (27)$$

where $p_{i,m,n,t}$ is the value per pound of the shipment and $\mathbf{X}_{i,t}$ are a set of controls such as destination, origin and quarter fixed effects. The estimates are reported in Table 5 in the Appendix and I use the estimate in the column (4).

Finally, I assume the economy is at a BGP in 2002 and set γ_L to 1.25% by targeting the average growth rate of the civilian labor force in the 10 years preceding my sample. For total population I use the sum of wage employment in that year from the sample in Section 2 so that $\bar{N} \approx 134$ mil. workers.

Internal Estimation For a given set Θ_2 , I estimate the remaining regional fundamentals $\{A_g, D_g\}$ by targeting output per worker and population shares in 2002 while solving the initial steady state. I use a modified version of Algorithm 1 where I solve for the fixed point of parameters in addition to endogenous variables. This also allows me to find λ and δ . I impute the latter by using equation 26 and the birth rates $\bar{b}_g^1$ observed 25 years before the first year in my sample. These are the initial elements in the series $\{b_g(t)\}$ I will use to simulate the transition. On the other hand, to infer the probability of migration opportunities I target the sum of migration inflows into all regions:

$$\underbrace{\frac{\sum_g \bar{m}_g}{\bar{N}}}_{\text{Observed}} = \lambda \cdot \underbrace{\sum_g \int \sum_{i \neq g} \bar{m}_{ig}(z) \cdot \bar{\psi}_i(z) dz}_{\text{Model Variable}} \quad (28)$$

Once this is complete, I compute the second endpoint for the transition: the final BGP. In terms of parameters, the only difference with the initial steady state is that I use the lagged birth rates corresponding to the last year of my sample $\bar{b}_g^2$.

With both endpoints in hand, I can now use Algorithm 2 to compute the transition dynamics of the economy. In particular, I leave all parameters constant except birth rates $\{b_g(t)\}$, which are the same as the ones I use to build the shift-share instrument. This results in a panel series of 17 years (including the first and last that I use for the BGP computation). Note it is unlikely the economy reaches the second steady state after this period. Therefore, I assume $b_g(t) = \bar{b}_g^2$ for all t thereafter and set the total periods to a higher number ($T = 200$).

To estimate Θ_2 , I follow an impulse response matching procedure. Specifically, I repeat the internal estimation steps described above for a sequence of parameters in a Sobol grid and seek to minimize the following objective function:

$$J = \min_{\Theta_2} (\hat{\Gamma} - \Gamma(\Theta_2))' W^{-1} (\hat{\Gamma} - \Gamma(\Theta_2)) \quad (29)$$

where $\hat{\Gamma}$ is a vector with the IRFs estimated using the 35 clusters, $\Gamma(\Theta_2)$ are the model generated IRFs. The weighting matrix W^{-1} is diagonal and contains the reciprocals of coefficient variance, similar to [Christiano, Eichenbaum and Evans \(2005\)](#).

5.2 Parameter Estimates

The results from the impulse response matching are displayed in Table 4 and Figure 7 shows the model fit.

TABLE 4 STRUCTURAL PARAMETERS

External Calibration				
Parameter	Description	Value	Source	
ρ	Discount Parameter	0.10	Crews (2022)	
σ	Elasticity of Substitution Across Regions	1.10	Boehm et al. (2023)	
$\bar{z}^f$	Mean of Initial Distribution	92.3	Hugget (2006)	
CV^f	Coeff. of Variation of Initial Distribution	0.481	Hugget (2006)	
Internal Calibration				
Parameter	Description	Mean/Estimate	Range	Moment
δ	Worker Exit Rate	8.37E-05	-	Population Growth in BGP 1
T_g	Regional Land	88646.72	[21752.93 ; 167221.74]	USGS geodata
A_g	Regional Productivity	9.73E-03	[8.08E-03 ; 1.12E-02]	Output per Worker in BGP 1
D_g	Regional Amenities	10.09	[0.64 ; 32.09]	Population Shares in BGP 1
Impulse Response Matching				
Parameter	Description	Estimate	Moment	
ε	Elasticity of Substitution Across Varieties	5.46	Response of Average Wages (Δw_{it})	
η	Inter-temporal Elasticity	0.19	Response of Average Wages (Δw_{it})	
ν	Migration Elasticity	1.41	Response of Migration Flows ($\Delta m_{it} \& \Delta o_{it}$)	
λ	Migration Opportunity Rate	0.03	Response of Migration Flows ($\Delta m_{it} \& \Delta o_{it}$)	
ϕ	Elasticity of Amenities to Population	1.74	Response of Migration Flows ($\Delta m_{it} \& \Delta o_{it}$)	
θ	Share of Land in Housing	0.17	Response of Housing Prices (Δhp_{it})	

6 Population and Relocation Shocks

How would aggregate output per worker respond over time if some regions lost a share of their workers? Although the IRFs in Section 3 show what the effect of migration on local conditions is, they are not enough to infer the relevance of population flows on aggregate variables. Therefore, to answer this question, I use the calibrated model. Specifically, I consider two policy exercises separately. The first is a “deportation” shock that decreases the total number of workers in the economy by reducing the population in certain areas while keeping the size of the remaining regions constant. The second one is a “relocation” shock: instead of maintaining the population of other locations, I increase it by a uniform share so as to keep the total number of households constant.

An important element of these counterfactuals is that the local population share that is removed or relocated differs across treated regions so that these local shocks are a geographically heterogeneous. To calibrate these shares I use the estimates from [Passel and Krogstad \(2023\)](#) to identify the states with the highest number of undocumented immigrants. I then match these with one of the 35 clusters in my model and perform the counterfactuals. Consequently, we can think of the first shock as a general deportation of approximately 1% of the labor force. By region however, this percentage varies from 3% to 5%. On the other hand, in the second exercise these workers are redistributed to places with a lower incidence of undocumented immigration, in a similar manner to the transportation of migrants to sanctuary cities by some states in 2023 (see for example [Goodman et al. \(2024\)](#)).

6.1 Local Deportations

Figure 8 displays the effects of the initial shock. It reveals that remaining residents in regions directly impacted by the deportation experience less congested amenities and an average increase in real incomes of 1.8%, after adjusting for prices of consumption goods and housing services. As these regions become more attractive, more workers choose to move there, resulting in an immediate 20% increase in gross inflows and a 4% decrease in outflows, while the opposite trends are observed in non-treated areas. Over time, as households choose to leave these non-treated regions, their amenities also become less congested.

Over time, the effects of the initial shock begin to dissipate, and most variables stabilize at new levels relative to the original steady state. With fewer workers in the economy, amenities in the long run increase by 2.5%, and real incomes rise by 1%. This aligns with the long-term effect on output, shown in Figure 9, which exhibits a decrease of almost 2%. Notably, if the shock were to target high-skilled individuals instead of being uniformly distributed across the regional productivity distribution $\psi_g(z, t)$, output would experience a larger decline, while real incomes would be higher. Intuitively, such a policy constitutes a labor supply shock that lower the average skill level available in affected regions, thereby

increasing wages real while lowering total output. Given the low estimated share θ of land in the production of housing services, the higher wages are passed-through as higher housing prices.

Local populations also converge to a new steady state that is 1% lower than the level observed before the policy was implemented. A smaller workforce implies reduced migration flows, as shown in Figure 8. Additionally, a lower labor supply means fewer firms can cover the fixed costs of operation, leading to a decrease in the number of available varieties. This effect is further amplified by the dynamic influence of $M_g(t)$ on fixed costs, as described in equation (9). Together, these mechanisms contribute to a steady decline in the mass of firms, illustrated in Figure 9. A similar process unfolds in unaffected areas, where population loss due to migration outflows also reduces labor and variety levels, ultimately decreasing output in those regions. Meanwhile, in the regions affected by the shock, migration inflows gradually increase the variety of goods over time.

In terms of aggregate effects, the deportation shock results in a long-run decline in output per worker of 0.5%, as shown in Figure 10. This new steady state arises because workers exit local economies without being replaced elsewhere, unlike the “relocation” shock I explore later. Consequently, the overall economy is left with a smaller labor supply and, therefore, fewer varieties, which reduces productivity. This outcome is similar to an inverse version of the refugee allocation scenario in Peters (2022), where output per worker also rises in the long run. However, in this case, the effect is driven by agglomeration externalities across all markets, rather than being limited to the manufacturing sector. Additionally, it is worth noting the different impacts depending on the type of workers removed. If high-skilled workers leave, aggregate output per capita immediately decreases to the new level. By contrast, if the shock removes low-skilled labor, output per capita initially rises, as the average skill level in the economy increases. Over time, however, the same dynamics that reduce varieties come into play, though at a slower rate.

6.2 Relocating Workers

How would aggregate productivity react if instead of removing workers they were re-allocated to other sectors? Effectively, this policy also takes the economy out of its initial steady state but does not change the BGP towards which the economy transitions to in the long-run. Figure 11 shows that, similar to before, regions that lose their population see an increase of inflows and a reduction of outflows, whereas the opposite is true of areas that receive the initial influx. Right after the shock, congested amenities and lower real incomes drive migrants back to areas whose population was relocated. This dynamic continues for 50 years until the number of workers in each places converges back.

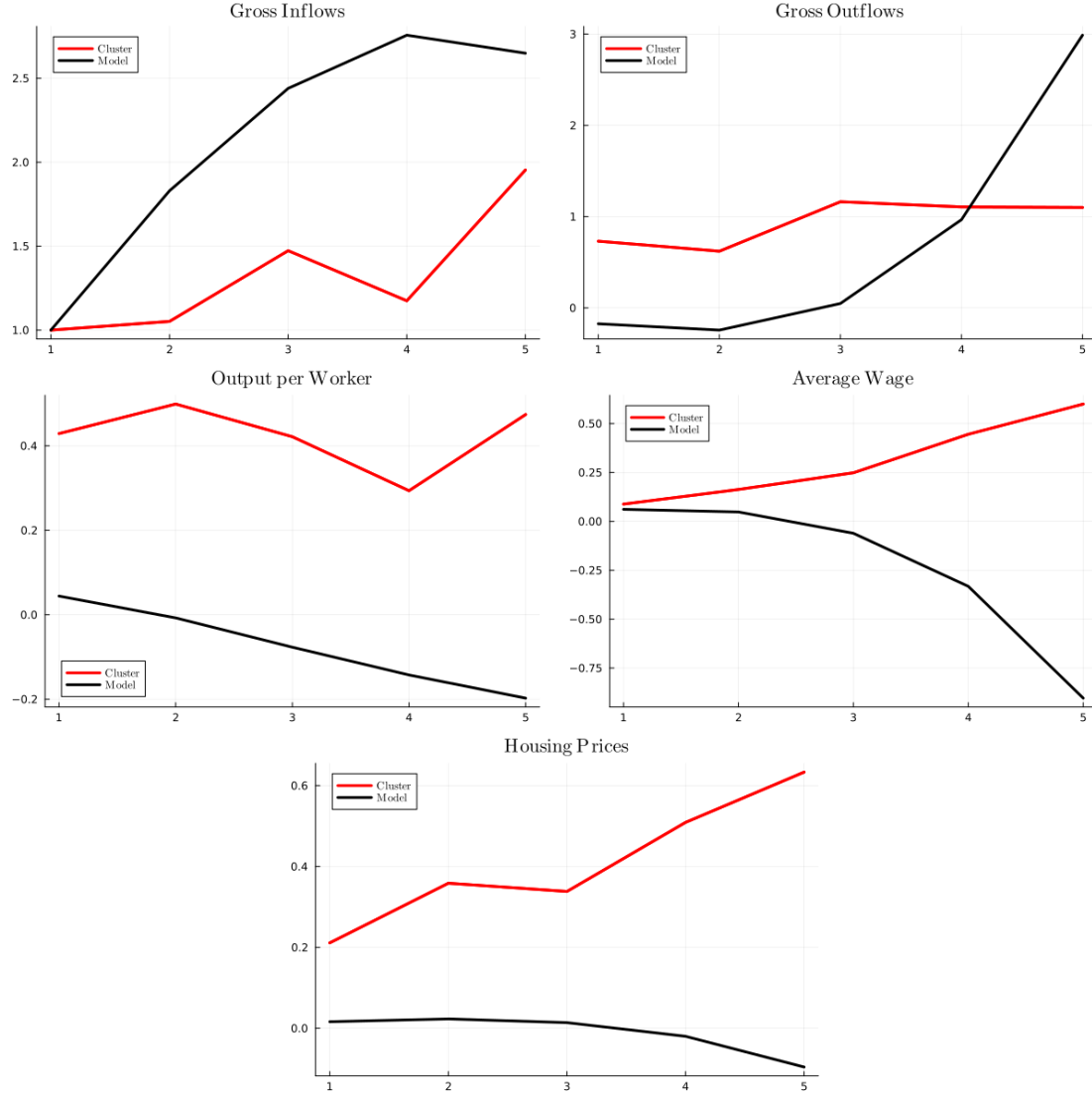
The mass of firms in each region also transition back to the initial steady state, as well as output. However, unlike population, these variables take longer to converge. The number

of producers is a slow moving variable that follows the differential equation 20. It therefore takes time to adjust, which is why the initial drop in shocked areas also occurs in a longer time frame relative to labor.

7 Conclusion

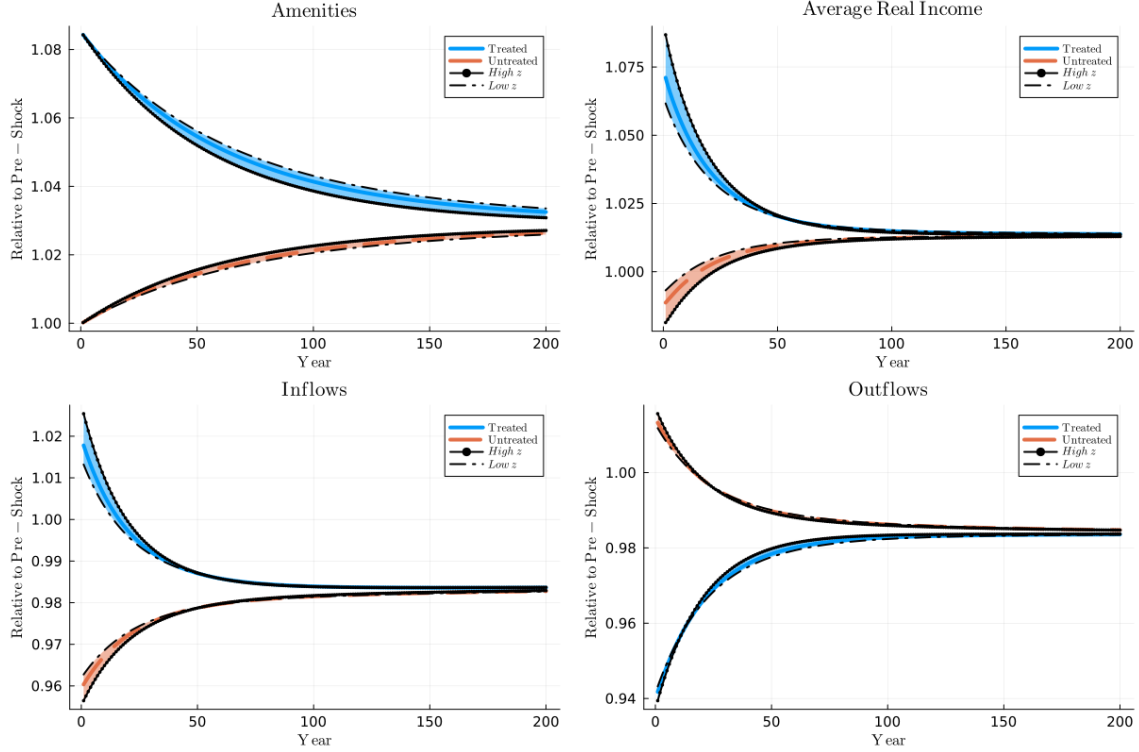
This paper studies the dynamic effect of increasing population inflows on local markets. Most of the previous research on labor mobility and its impact on economic outcomes focused on either short or long-term responses, but did not consider transitional dynamics. Although some papers address this element, their identification of structural migration shocks relies on timing assumptions, whereas the analysis above adopts an SVAR-IV approach using instruments based on weather variations. The results reveal labor productivity, wages and employment opportunities in urban economies respond positively to a 1% increase in population inflows.

FIGURE 7. MODEL FIT



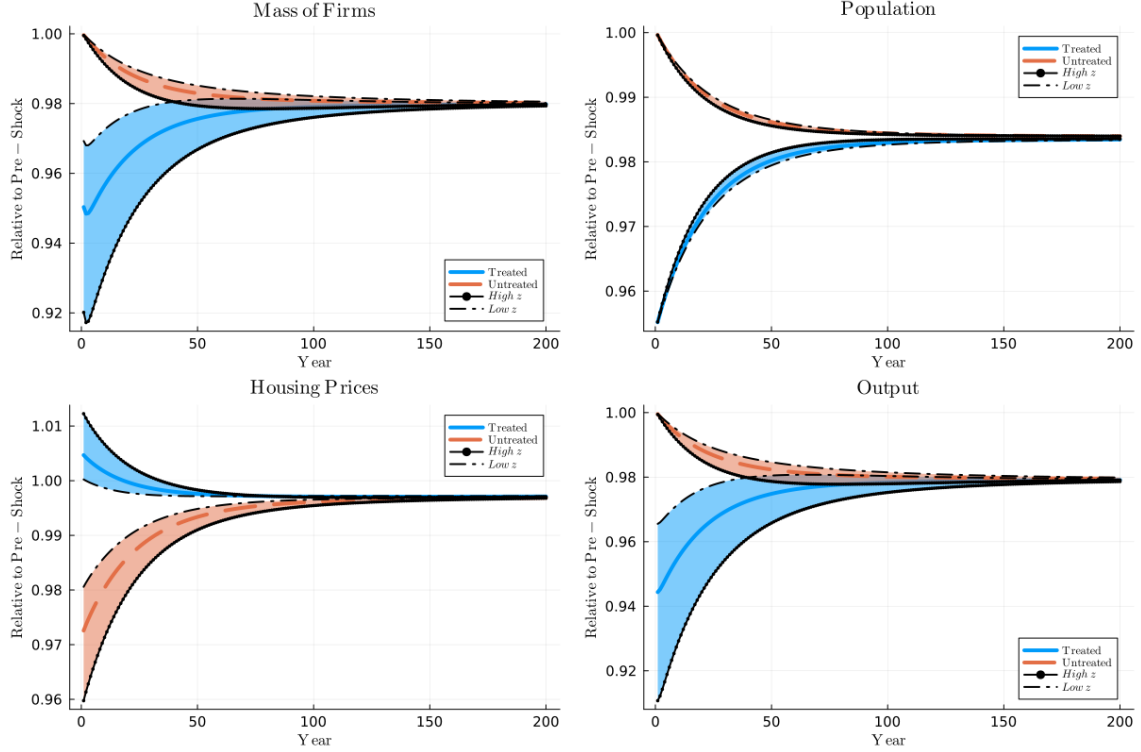
Note: The red line represents the cumulative impulse response (CIRF) estimated from the 35 K-means clusters. The red shade indicates a 90% confidence level. Standard errors are generated by Monte-Carlo with 200 repetitions. The blue line is the CIRF estimated by simulating the model with the baseline parametrization obtained from indirect inference.

FIGURE 8. DEPORTATION SHOCK: UTILITY AND POPULATION FLOWS



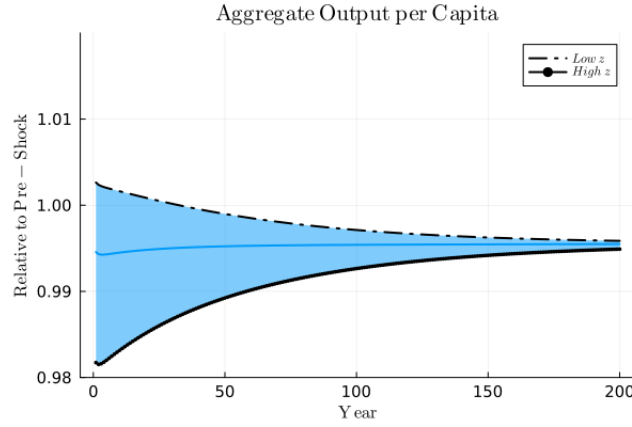
Note: Responses of amenities, real income, and gross migration flows to the “deportation” shock where heterogeneous regional population shares are removed from the economy. The blue line represents the response in regions that experience a population reduction (“treated”) whereas the red line represents the effect in the remaining ones (“untreated”). In both settings, the policy removes a uniform share across the regional productivity distribution, whereas the bands represent the cases where only the top 10% (“High z ”) or bottom 10% (“Low z ”) are removed. Variables x_t are plotted relative to their pre-shock levels $\hat{x}_t = x_t/x_{pre-shock}$.

FIGURE 9. DEPORTATION SHOCK: POPULATION, VARIETIES AND OUTPUT



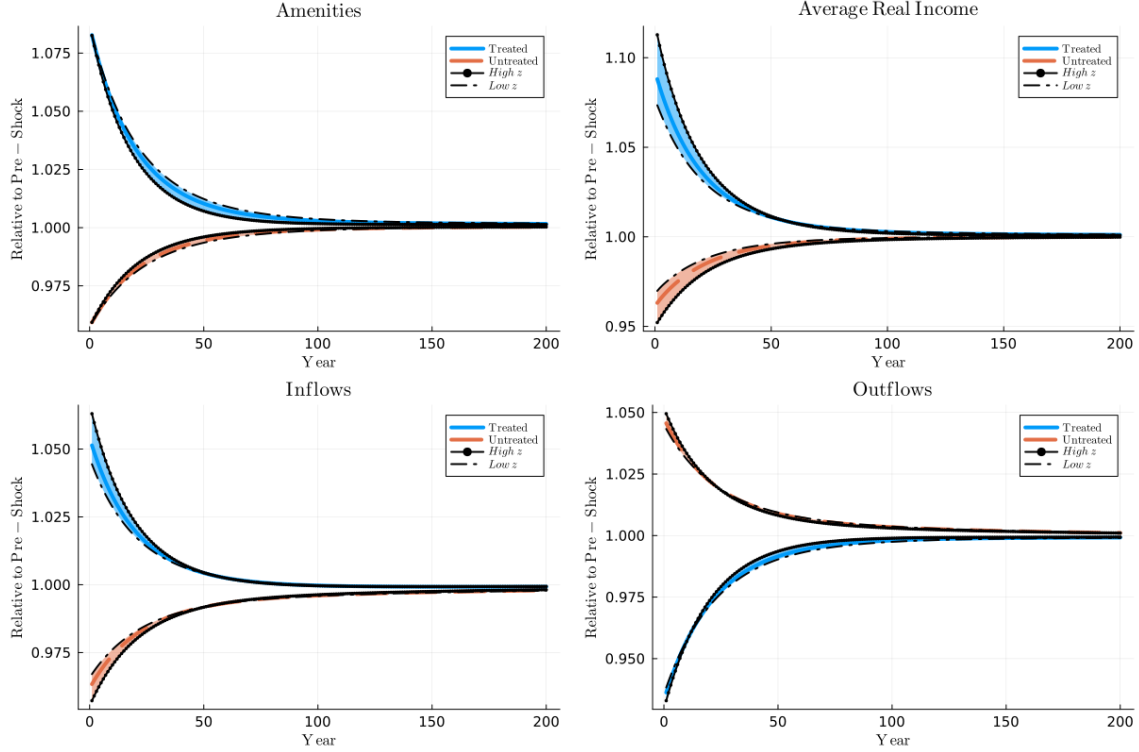
Note: Responses of varieties, population, housing prices and output to the “deportation” shock where heterogeneous regional population shares are removed from the economy. The blue line represents the response in regions that experience a population reduction (“treated”) whereas the red line represents the effect in the remaining ones (“untreated”). In both settings, the policy removes a uniform share across the regional productivity distribution, whereas the bands represent the cases where only the top 10% (“High z ”) or bottom 10% (“Low z ”) are removed. Variables x_t are plotted relative to their pre-shock levels $\hat{x}_t = x_t/x_{pre-shock}$.

FIGURE 10. DEPORTATION SHOCK: AGGREGATE OUTPUT



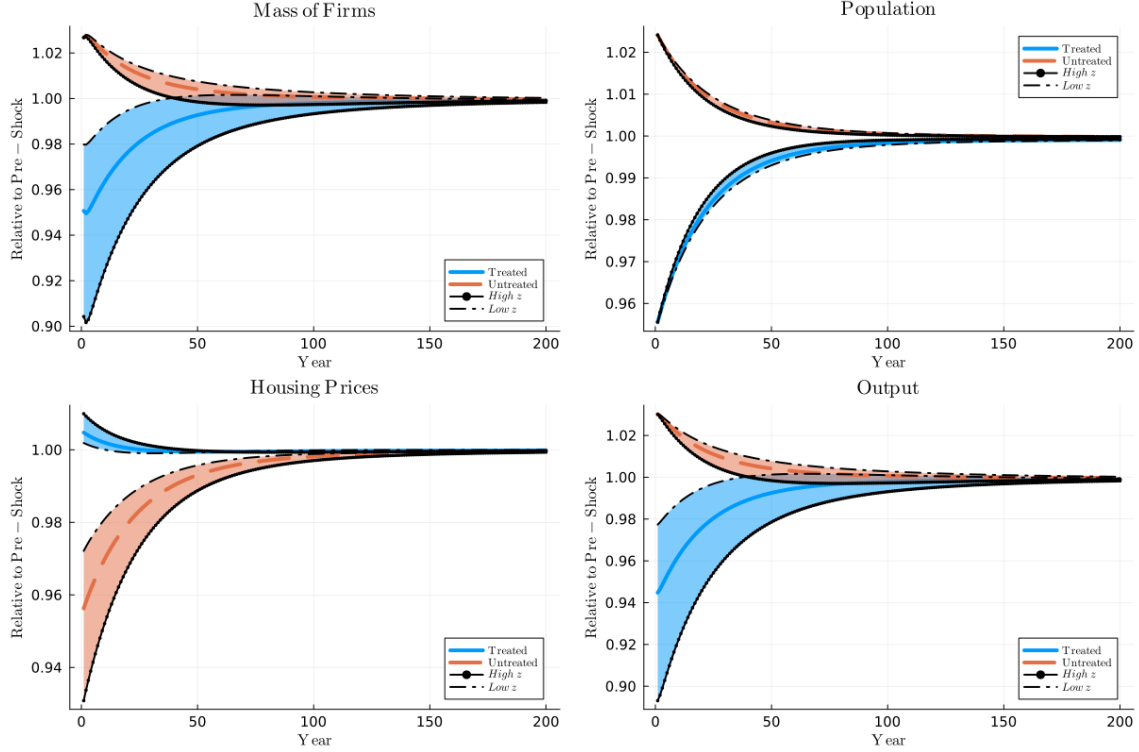
Note: Response of aggregate output per worker to the “deportation” shock where heterogeneous regional population shares are removed from the economy. The blue line represents the response when the shock removes a uniform share across the regional productivity distribution, whereas the bands represent the cases where only the top 10% (“High z ”) or bottom 10% (“Low z ”) are removed. Output per worker is plotted relative to its pre-shock levels $\hat{x}_t = x_t/x_{pre-shock}$.

FIGURE 11. RELOCATION SHOCK: UTILITY AND POPULATION FLOWS



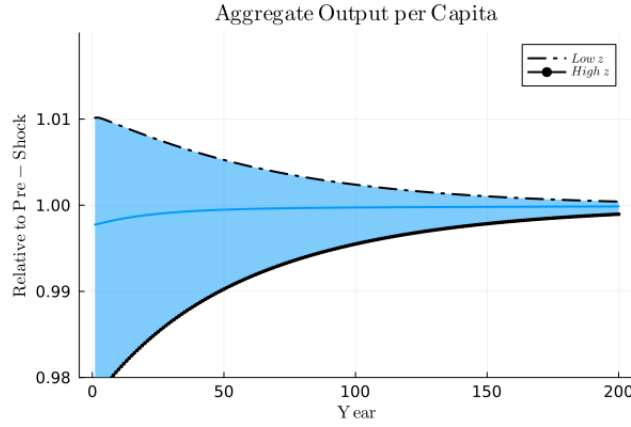
Note: Responses of amenities, real income, and gross migration inflows to the “deportation” shock where heterogeneous regional population shares are removed from the economy. The blue line represents the response in regions that experience a population reduction (“treated”) whereas the red line represents the effect in the remaining ones (“untreated”). In both settings, the policy removes a uniform share across the regional productivity distribution, whereas the bands represent the cases where only the top 10% (“High z ”) or bottom 10% (“Low z ”) are removed. Variables x_t are plotted relative to their pre-shock levels $\hat{x}_t = x_t/x_{pre-shock}$.

FIGURE 12. RELOCATION SHOCK: POPULATION, VARIETIES AND OUTPUT



Note: Responses of varieties, population, housing prices and output to the “deportation” shock where heterogeneous regional population shares are removed from the economy. The blue line represents the response in regions that experience a population reduction (“treated”) whereas the red line represents the effect in the remaining ones (“untreated”). In both settings, the policy removes a uniform share across the regional productivity distribution, whereas the bands represent the cases where only the top 10% (“High z ”) or bottom 10% (“Low z ”) are removed. Variables x_t are plotted relative to their pre-shock levels $\hat{x}_t = x_t/x_{pre-shock}$.

FIGURE 13. RELOCATION SHOCK: AGGREGATE OUTPUT



Note: Response of aggregate output per worker to the “deportation” shock where heterogeneous regional population shares are removed from the economy. The blue line represents the response when the shock removes a uniform share across the regional productivity distribution, whereas the bands represent the cases where only the top 10% (“High z ”) or bottom 10% (“Low z ”) are removed. Output per worker is plotted relative to its pre-shock levels $\hat{x}_t = x_t/x_{pre-shock}$.

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A Theory Appendix

A.1 Dynamic Equilibrium Solution

A.1.1 Derivation of the HJB

Following similar steps to those in [Crews \(2023\)](#) we find the following Bellman in discrete time:

$$V_g(z, t) = U_g(z, t) + [(1 - \rho)(1 - \delta)] \mathbb{E}_t \left[\tilde{\xi}(t+1) \tilde{\kappa}(t+1) V_{g_{t+1}}(z, t+1) \right]$$

The expectation is with respect to the location in the next period since the households does not know whether or not it will have the opportunity to move (which occurs with probability λ), and what the preference shock will be. Thus we can rewrite the equation above as:

$$V_g(z, t) = U_g(z, t) + [(1 - \rho)(1 - \delta)] \left\{ (1 - \lambda) V_g(z, t+1) + \lambda \mathbb{E}_{\xi_i} \left[\max_i \xi_i \kappa_{g,i} V_i(z, t+1) \right] \right\}$$

To simplify notation define $V_{gi,t+1} = \kappa_{g,i} V_i(z, t+1)$. Since ξ_i follows a Frechet distribution with shape parameter ν and scale parameter¹⁰ $s = \Gamma_\nu^{-1}$, we can find an expression for:

$$\begin{aligned} \tilde{F}(y) &= P \left(\max_i \{ \xi_i V_{gi,t+1} \} \leq y \right) \\ &= P \left(\xi_1 \cdot V_{g1,t+1} \leq y \cap \dots \cap \xi_G \cdot V_{gG,t+1} \leq y \right) \\ &= P \left(\xi_1 \leq \frac{y}{V_{g1,t+1}} \right) \dots P \left(\xi_G \leq \frac{y}{V_{gG,t+1}} \right) \\ &= \exp \left(- \left(\frac{y}{s \cdot V_{g1,t+1}} \right)^{-\nu} \right) \dots \exp \left(- \left(\frac{y}{s \cdot V_{gG,t+1}} \right)^{-\nu} \right) \\ &= \exp \left(- \left(\frac{y}{s} \right)^{-\nu} \cdot \sum_{i=1}^G (V_{gi,t+1})^\nu \right) \end{aligned}$$

Notice we can rewrite this as:

$$\tilde{F}(y) = \exp \left(- \left[\frac{y}{s \cdot \Phi} \right]^{-\nu} \right)$$

where $\Phi = [\sum_{i=1}^G (V_{gi,t+1})^\nu]^\frac{1}{\nu}$. Thus $y \sim \text{Frechet}(s \cdot \Phi, \nu)$ and therefore:

$$\begin{aligned} E_{\xi_i} [\max_i \xi_i V_{gi,t+1}] &= s \cdot \Phi \cdot \Gamma_\nu \\ &= \left[\sum_{i=1}^G (V_{gi,t+1})^\nu \right]^\frac{1}{\nu} \end{aligned}$$

¹⁰Where $\Gamma_\nu = \Gamma(1 - \frac{1}{\nu})$.

where $\Gamma_\nu = \Gamma\left(1 - \frac{1}{\nu}\right)$. Also note that, for any destination $d \in G$ (and dropping t for simplicity), migration flows are given by:

$$m_{gd}(z, t+1) = \frac{[V_{gd,t+1}]^\nu}{\sum_i [V_{gi,t+1}]^\nu}$$

So the expectation above becomes:

$$E_{\xi_i}[\max_i \xi_i V_{gi,t+1}] = \left[\frac{[V_{gd,t+1}]^\nu}{m_{gd}(z, t+1)} \right]^{\frac{1}{\nu}} = m_{gd}(z, t+1)^{-\frac{1}{\nu}} \cdot V_{gd,t+1}$$

Note that, since this applies $\forall d \in G$, we can express the expectation as:

$$E_{\xi_i}[\max_i \xi_i V_{gi,t+1}] = \frac{1}{G} \sum_i m_{gi}(z, t+1)^{-\frac{1}{\nu}} \cdot V_{gi,t+1}$$

Hence the Bellman in discrete time is given by:

$$V_g(z, t) = U_g(z, t) + [(1 - \rho)(1 - \delta)] \left\{ (1 - \lambda)V_g(z, t+1) + \lambda \frac{1}{G} \sum_i m_{gi}(z, t+1)^{-\frac{1}{\nu}} \cdot \kappa_{gi} V_i(z, t+1) \right\}$$

To convert to continuous time, consider a period of length Δ to the Bellman becomes:

$$V_g(z, t) = U_g(z, t) + [(1 - \Delta\rho)(1 - \Delta\delta)] \left\{ (1 - \Delta\lambda)V_g(z, t + \Delta) + \Delta\lambda \frac{1}{G} \sum_i m_{gi}(z, t + \Delta)^{-\frac{1}{\nu}} \cdot \kappa_{gi} V_i(z, t + \Delta) \right\}$$

After multiplying everything, subtracting $V_g(z, t)$, dividing by Δ and taking the limit $\Delta \rightarrow 0$ we obtain the HJB in equation (16):

$$(\delta + \rho + \lambda)V_g(z, t) = U_g(z, t) + \dot{V}_g + \lambda \left[\frac{1}{G} \sum_i m_{gi}(z, t)^{-\frac{1}{\nu}} \cdot \kappa_{gi} V_i(z, t) \right]$$

A.1.2 Derivation of the Kolmogorov forward equation

By definition the distribution of efficiency units z at any point in time t and region g is given by:

$$\Psi_g(z, t) = \mathbb{P}\{Z \leq z, g(t) = g\}$$

Therefore, the number of people in region g with efficiency units less than or equal to z will be given by $N_g(z, t) = \Psi_g(z, t)N(t)$, so that:

$$\dot{N}_g(z, t) = \dot{\Psi}_g(z, t) \cdot N(t) + \Psi_g(z, t) \cdot \dot{N}(t)$$

On the other hand, in discrete time, after each interval of length Δ , $E_g(t)$ workers are born each period, which draw their skills from the distribution $F(z)$. Furthermore, a share of $(1 - \delta)$ households remain alive in $t + \Delta$. Of those, λ will not be able to move whereas $(1 - \lambda)$ from other regions can migrate to g from $i \in G$, and do so with probability $m_{ig}(z, t + \Delta)$. Consequently, the total share of workers with skill z moving from i to g is given by $\Delta \lambda m_{ig}(z, t + \Delta)$.

Combining all these elements, we obtain the following evolution of $N_g(z, t)$:

$$N_g(z, t + \Delta) = \Delta E_g(t) \cdot F(z) + (1 - \Delta \delta) \left[(1 - \Delta \lambda) N_g(z, t) + \Delta \lambda \sum_i m_{ig}(z, t + \Delta) N_i(z, t) \right]$$

Subtracting $N_g(z, t)$, dividing by Δ and taking the limit $\Delta \rightarrow \infty$ we have:

$$\dot{N}_g(z, t) = E_g(t) \cdot F(z) - \delta N_g(z, t) - \lambda \left[(1 - m_{gg}(z, t)) N_g(z, t) - \sum_{i \neq g} m_{ig}(z, t) N_i(z, t) \right]$$

where $\dot{\Psi} = \partial_t \Psi$. Finally, combine both expressions for $\dot{N}_g(z, t)$ and take derivatives with respect to z to obtain the Kolmogorov forward equation in equation (18):

$$\begin{aligned} \dot{\psi}_g(z, t) &= \frac{E_g(t)}{N(t)} \cdot f(z) - \delta \psi_g(z, t) \\ &- \lambda \left[(1 - m_{gg}(z, t)) \psi_g(z, t) - \sum_{i \neq g} m_{ig}(z, t) \psi_i(z, t) \right] \\ &- \psi_g(z, t) \frac{\dot{N}(t)}{N(t)} \end{aligned}$$

where $\bar{z}$ is the maximum productivity level.

A.1.3 Detrended Variables and Balanced Growth Path

For any variable $y(t)$, denote the detrended version as $\tilde{y}(t) = e^{-\gamma_y t} y(t)$, where γ_y is the growth rate of y , and in the case of population, set $N(0) = N_0$ as its initial condition. Also, let $\bar{y}$ be the same variable along the balanced growth path. In a BGP, the productivity distribution is stationary and the mass of intermediate firms grows at a common constant rate $\gamma_M = \dot{M}_g(t)/M_g(t)$. In this case, the evolution of $M_g(t)$ implies $\gamma_M = F_g(t) \cdot M_g(t)^\eta$ and therefore the growth rate γ_L is determined by combining this with the production labor demand equation:

$$\begin{aligned} \gamma_M &= F_g(t) \cdot M_g(t)^\eta \\ \gamma_M &= \left(\frac{L_g^y(t)}{\varepsilon M_g(t)} \right) M_g(t)^\eta \\ \gamma_M &= \left(\frac{1}{\varepsilon} \right) \tilde{L}_g^y(t) \cdot (\tilde{M}_g(t))^{\eta-1} e^{[\gamma_L + \gamma_M(\eta-1)]t} \end{aligned}$$

which implies $\gamma_L = \gamma_M(1 - \eta)$. Recall from the labor market clearing condition:

$$L_g(t) = L_g^y(t) + L_g^h(t)$$

so that:

$$L_g^y(t) = 1 - (1 - \theta)(1 - \alpha)L_g(t) = [\alpha + \theta(1 - \alpha)]L_g(t)$$

So that:

$$\gamma_M = \left(\frac{\alpha + \theta(1 - \alpha)}{\varepsilon} \right) L_g(t) \cdot (\tilde{M}_g(t))^{\eta-1}$$

From the definition of $L_g(t)$ we can find γ_L :

$$\begin{aligned} L_g(t) &= \left(N(t) \cdot \int_{\Omega_z} z^\beta \psi_g(z, t) dz \right)^{\frac{1}{\beta}} \\ \tilde{L}_g(t) \cdot e^{\gamma_L t} &= e^{\frac{1}{\beta}(\gamma_N + \gamma_\psi)t} \left(\tilde{N}(t) \cdot \int_{\Omega_z} z^\beta \psi_g(z, t) dz \right)^{\frac{1}{\beta}} \end{aligned}$$

which implies $\gamma_L = \frac{1}{\beta}\gamma_N$ (we will see later that $\gamma_\psi = 0$). We can use the solution for $U_g(\cdot)$ from the static equilibrium to find detrended flow utility:

$$\begin{aligned} U_g(z, t) &= D_g(t) \cdot z^\beta \cdot \frac{\Gamma_g}{\Gamma_U} \cdot \left(\pi_{gg}(t) \right)^{\frac{\alpha}{1-\sigma}} \left(M_g(t) \right)^{\frac{\alpha}{\varepsilon-1}} \left(L_g(t) \right)^{(1-\beta)-\theta(1-\alpha)} \\ \tilde{U}_g(z, t) e^{\gamma_U t} &= e^{[\gamma_M \frac{\alpha}{\varepsilon-1} + \gamma_L((1-\beta)-\theta(1-\alpha)) - \gamma_N \phi]t} \\ &\quad \cdot D_g \tilde{N}_g(t)^{-\phi} z^\beta \cdot \frac{\Gamma_g}{\Gamma_U} \cdot \left(\pi_{gg}(t) \right)^{\frac{\alpha}{1-\sigma}} \left(\tilde{M}_g(t) \right)^{\frac{\alpha}{\varepsilon-1}} \left(\tilde{L}_g(t) \right)^{(1-\beta)-\theta(1-\alpha)} \end{aligned}$$

which, along with the expressions $\gamma_N = \beta\gamma_L$ and $\gamma_L = \gamma_M(1 - \eta)$, implies:

$$\gamma_U = \left[\frac{\alpha}{\varepsilon-1} - (1 - \eta)(\beta(1 + \phi) + \theta(1 - \alpha) - 1) \right] \gamma_M$$

Detrended Equations For the value function, recall that by definition in a BGP, $V_g(z, t) = e^{\gamma_U t} \tilde{V}_g(z, t)$. To find the detrended HJB, we can then rewrite each element in the HJB:

$$\begin{aligned} (\delta + \rho)V_g(z, t) &= (\delta + \rho)e^{\gamma_U t} \tilde{V}_g(z, t) \\ U_g(z, t) &= e^{[\frac{\alpha}{\varepsilon-1} - (1-\eta)(\beta(1+\phi) + \theta(1-\alpha) - 1)]\gamma_M t} \tilde{U}_g(z, t) \\ \partial_t V_g(z, t) &= \gamma_U e^{\gamma_U t} \tilde{V}_g(z, t) + e^{\gamma_U t} \partial_t \tilde{V}_g(z, t) \end{aligned}$$

Combining all this into the HJB we obtain:

$$\begin{aligned} (\delta + \rho)e^{\gamma_U t} \tilde{V}_g(z, t) &= e^{[\frac{\alpha}{\varepsilon-1} - (1-\eta)(\beta(1+\phi) + \theta(1-\alpha) - 1)]\gamma_M t} \tilde{U}_g(z, t) \\ &\quad + \gamma_U e^{\gamma_U t} \tilde{V}_g(z, t) + e^{\gamma_U t} \partial_t \tilde{V}_g(z, t) \\ &\quad + e^{\gamma_U t} \lambda \sum_i m_{gi}(z, t) \left[\delta_{gi}(z, t) \tilde{V}_i(z, t) - \tilde{V}_g(z, t) \right] \end{aligned}$$

This implies:

$$\gamma_v = \left(\frac{\alpha}{\varepsilon-1} - (1-\eta)(\beta(1+\phi) + \theta(1-\alpha) - 1) \right) \gamma_M$$

which results in the detrended HJB:

$$(\delta + \rho - \gamma_v) \tilde{V}_g(z, t) = \tilde{U}_g(z, t) + \partial_t \tilde{V}_g(z, t) + \lambda \sum_i m_{gi}(z, t) \left[\delta_{gi}(z, t) \tilde{V}_i(z, t) - \tilde{V}_g(z, t) \right]$$

For the detrended Kolmogorov equation consider each element of the equation:

$$\partial_t \psi_g = \gamma_\psi e^{\gamma_\psi t} \tilde{\psi}_g(z, t) + e^{\gamma_\psi t} \partial_t \tilde{\psi}_g(z, t)$$

$$\frac{E_g(t)}{N(t)} f(z) - \delta \psi_g(z, t) = \frac{\tilde{E}_g(t)}{\tilde{N}(t)} f(z) - \delta \tilde{\psi}_g(z, t) e^{\gamma_\psi t}$$

$$\lambda(1 - m_{gg}(z, t)) \psi_g(z, t) = \lambda(1 - m_{gg}(z, t)) \tilde{\psi}_g(z, t) e^{\gamma_\psi t}$$

$$\lambda \sum_{i \neq g} m_{ig}(z, t) \psi_i(z, t) = \lambda \sum_{i \neq g} m_{ig}(z, t) \tilde{\psi}_i(z, t) e^{\gamma_\psi t}$$

$$\frac{\dot{N}(t)}{N(t)} \psi_g(z, t) = \left(\tilde{\gamma}_N(t) + \gamma_N \right) \tilde{\psi}_g(z, t) e^{\gamma_\psi t}$$

where $\tilde{\gamma}_N(t) = \dot{\tilde{N}}(t)/\tilde{N}(t)$ is the growth rate of the detrended total population. We can obtain this result since $N(t) = \tilde{N}(t)e^{\gamma_N t}$, which implies:

$$\dot{N}(t) = \frac{dN(t)}{dt} = \frac{d\tilde{N}(t)}{dt} \cdot e^{\gamma_N t} + e^{\gamma_N t} \gamma_N \tilde{N}(t)$$

so that:

$$\frac{\dot{N}(t)}{N(t)} = \frac{\dot{\tilde{N}}(t)}{\tilde{N}(t)} + \gamma_N$$

Combining them all we obtain:

$$\begin{aligned} \gamma_\psi e^{\gamma_\psi t} \tilde{\psi}_g(z, t) + e^{\gamma_\psi t} \partial_t \tilde{\psi}_g(z, t) &= \frac{\tilde{E}_g(t)}{\tilde{N}(t)} f(z) - \delta \tilde{\psi}_g(z, t) e^{\gamma_\psi t} \\ &- \lambda[1 - m_{gg}(z, t)] \tilde{\psi}_g(z, t) e^{\gamma_\psi t} \\ &+ \lambda \sum_{i \neq g} m_{ig}(z, t) \tilde{\psi}_i(z, t) e^{\gamma_\psi t} - \left(\tilde{\gamma}_N(t) + \gamma_N \right) \tilde{\psi}_g(z, t) e^{\gamma_\psi t} \end{aligned}$$

which implies that $\gamma_\psi = 0$. Thus, using this, we obtain the detrended Kolmogorov equation:

$$\begin{aligned}\partial_t \tilde{\psi}_g(z, t) &= \frac{\tilde{E}_g(t)}{\tilde{N}(t)} f(z) - \delta \tilde{\psi}_g(z, t) \\ &- \lambda \left[(1 - m_{gg}(z, t)) \tilde{\psi}_g(z, t) - \sum_{i \neq g} m_{ig}(z, t) \tilde{\psi}_i(z, t) \right] - \left(\tilde{\gamma}_N(t) + \gamma_N \right) \tilde{\psi}_g(z, t)\end{aligned}$$

Detrended growth Note that, using population dynamics, we can derive the detrended growth rate. By definition:

$$\dot{N}(t) = \sum_g (E_g(t) - \delta \cdot N_g(t))$$

Therefore:

$$\frac{\dot{N}(t)}{N(t)} = \frac{\sum_g (E_g(t) - \delta \cdot N_g(t))}{N(t)} = \frac{\sum_g \tilde{E}_g(t)}{\tilde{N}(t)} - \delta$$

where the second equality follows from $E_g(t) = e^{\gamma_N t} \tilde{E}_g(t)$ and $N(t) = \sum_g N_g(t)$. Therefore, combining this with equation above we have:

$$\frac{\dot{\tilde{N}}(t)}{\tilde{N}(t)} = \frac{\sum_g \tilde{E}_g(t)}{\tilde{N}(t)} - \delta - \gamma_N$$

BGP Equations In the BGP we have $\tilde{y}(t) = \bar{y}$. Thus the HJB and Kolmogorov forward equations become:

$$\begin{aligned}(\delta + \rho - \gamma_v) \bar{V}_g(z) &= \bar{U}_g(z) + \lambda \sum_i \bar{m}_{gi}(z) \left[\bar{\delta}_{gi}(z) \bar{V}_i(z) - \tilde{V}_g(z) \right] \\ 0 &= \frac{\bar{E}_g}{\bar{N}} f(z) - (\delta + \gamma_N) \bar{\psi}_g(z) - \lambda (1 - \bar{m}_{gg}(z)) \bar{\psi}_g(z) + \lambda \sum_{i \neq g} \bar{m}_{ig}(z) \bar{\psi}_i(z)\end{aligned}$$

Note that along the BGP there is no net migration so that:

$$\sum_g \int \lambda [1 - \bar{m}_{gg}(z)] \bar{\psi}_g(z) dz = \sum_g \int \lambda \sum_{i \neq g} \bar{m}_{ig}(z) \bar{\psi}_i(z) dz$$

Also recall that $N(t) \Psi_g(\bar{z}) = N(t) \int \psi_g(z, t) = N_g(t)$ so that $\bar{\Psi}_g(\bar{z}) = \bar{N}_g / \bar{N}$. Thus, to derive the equilibrium population growth rate γ_N , integrate the BGP KF equation over z

and sum across regions g to obtain:

$$\begin{aligned}
0 &= \sum_g \frac{\bar{E}_g}{\bar{N}} - \sum_g \delta \int \bar{\psi}_g(z) dz - \gamma_N \\
\gamma_N &= \sum_g \frac{\bar{E}_g}{N_0} - \sum_g \delta \frac{\bar{N}_g}{N_0} \\
\gamma_N &= \sum_g \frac{\bar{E}_g}{N_0} - \delta
\end{aligned}$$

where we also use the fact that $\int f(z) dz = 1$ and imposing $\bar{N} = N_0$.

B Computation Appendix

Algorithm 1: Balanced Growth Path

Data: parameters $\{\Theta, \Theta_G\}$; productivity distribution $f(z)$

Result: solution functions $\{\bar{V}, \bar{\psi}, \bar{m}\}$

Initialize $\{\mathbf{W}^0, \mathbf{L}^0, \mathbf{N}^0\}$;

while *not converged* **do**

 Use $\{\mathbf{W}^n, \mathbf{L}^n, \mathbf{N}^n\}$ to compute shares π_{gd}^n and flow utility $\bar{U}_g^n(z)$;

 Solve the HJB in equation (24). Recover migration shares $\bar{m}_{gd}^n(z)$;

 Use $\bar{m}_{gd}^n(z)$ to solve the Kolmogorov forward equation (25). Recover the distribution $\bar{\psi}_g^n(z)$;

 Use $\bar{\psi}_g^n(z)$ to compute labor supply and population:

$$\hat{L}_g^n = N \cdot \int_{\Omega_z} z \bar{\psi}_g^n(z) dz \quad \hat{N}_g^n = N \cdot \int_{\Omega_z} \bar{\psi}_g^n(z) dz$$

 then use $\{\hat{L}^n\}$ to compute the wages that satisfy:

$$\hat{W}_g^n \hat{L}_g^n = \sum_d \pi_{gd}^n \hat{W}_d^n \hat{L}_d^n$$

 this is the trade balance in equation (12) ;

if $\{\mathbf{W}^n, \mathbf{L}^n, \mathbf{N}^n\}$ *close to* $\{\hat{\mathbf{W}}^n, \hat{\mathbf{L}}^n, \hat{\mathbf{N}}^n\}$ **then**

 converged;

else

 compute $\{\mathbf{W}^{n+1}, \mathbf{L}^{n+1}, \mathbf{N}^{n+1}\}$ as linear combination of old and new values;

end

end

Algorithm 2: Transition Dynamics

Data: endpoint solutions $\{\bar{V}^1, \bar{\psi}^1, \bar{m}^1\}$ and $\{\bar{V}^2, \bar{\psi}^2, \bar{m}^2\}$; parameter series $\{\Theta(t), \Theta_G(t)\}$; productivity distribution $f(z)$

Result: solution functions $\{\tilde{V}(t), \tilde{\psi}(t), m(t)\}$

Initialize $\{\mathbf{W}^0(t), \mathbf{L}^0(t), \mathbf{N}^0(t)\}$;

while *not converged* **do**

 Use $\{\mathbf{N}^j(t)\}$ to compute total population $\hat{N}^j(t)$;

 Use $\{\mathbf{L}^j(t)\}$ to compute a transition path $\{\hat{\mathbf{M}}^j(t)\}$ according to equation (20) using the initial condition from the first endpoint;

 Use $\{\mathbf{W}^j(t), \mathbf{L}^j(t), \mathbf{N}^j(t)\}$ and $\{\hat{\mathbf{M}}^j(t)\}$ to compute shares $\pi_{gd}^j(t)$ and flow utility $\tilde{U}_g^j(t)$;

 Solve the HJB equation (21) by iterating backwards from $\bar{V}^2$. Recover migration shares $m_{gd}^j(z, t)$;

 Use $m_{gd}^j(z, t)$ to solve the Kolmogorov forward equation (22) by iterating forward from $\bar{\psi}^1$. Recover $\psi_g^j(z, t)$;

 Use $\tilde{\psi}_g^j(z, t)$ to compute labor supply and population for each t :

$$\hat{L}_g^j(t) = \hat{N}^j(t) \cdot \int_{\Omega_z} z \cdot \tilde{\psi}_g^j(z, t) dz \quad \hat{N}_g^n = \hat{N}^j(t) \cdot \int_{\Omega_z} \tilde{\psi}_g^n(z, t) dz$$

 then use $\{\hat{\mathbf{L}}^j(t)\}$ to compute wages that satisfy:

$$\hat{W}_g^j(t) \hat{L}_g^n(t) = \sum_d \pi_{gd}^j \hat{W}_d^j(t) \hat{L}_d^j(t)$$

 this is the detrended trade balance equation.;

if $\{\mathbf{W}^j(t), \mathbf{L}^j(t), \mathbf{N}^j(t)\}$ *close to* $\{\hat{\mathbf{W}}^j(t), \hat{\mathbf{L}}^j(t), \hat{\mathbf{N}}^j(t)\}$ **then**

 converged;

else

 compute $\{\mathbf{W}^{j+1}(t), \mathbf{L}^{j+1}(t), \mathbf{N}^{j+1}(t)\}$ as linear combination of old and new values;

end

end

C Estimation Appendix

TABLE 5 REGRESSION OF PRICES ON DISTANCE

	$\log p_{i,m,n,j,t}$			
	(1)	(2)	(3)	(4)
ζ_τ	0.283*** (0.001)	0.169*** (0.001)	0.283*** (0.001)	0.168*** (0.001)
Same State	-0.467*** (0.002)	-0.210*** (0.002)	-0.467*** (0.002)	-0.211*** (0.002)
Orig. State	✓			
Dest. State	✓			
Orig. State * Prod.		✓		
Dest. State * Prod.		✓		
Orig. State * Quarter			✓	
Dest. State * Quarter			✓	
Orig. State * Prod. * Quarter				✓
Dest. State * Prod. * Quarter				✓
N	10,526,115	10,526,114	10,526,115	10,525,995
R^2	0.114	0.396	0.115	0.400